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AIUB, math 2 note

#Final

Term

~~Math 2~~
~~Final~~
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Topic Name : _____

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Multiple
Integrals
Ch 15

Formulas

m.o.m

$$\textcircled{1} \iint dx \cdot dy = \iint dA$$

$$\textcircled{2} \iint dA = \iint dx \cdot dy = \iint dy \cdot dx$$

$$\textcircled{3} \iiint dv = \iiint dx \cdot dy \cdot dz = \iiint dx \cdot dz \cdot dy$$

$$= \iiint dy \cdot dx \cdot dz = \iiint dy \cdot dz \cdot dx$$

$$= \iiint dz \cdot dx \cdot dy = \iiint dz \cdot dy \cdot dx$$

Example

m.o.m

$$* \int_0^2 \int_1^{x^2} dA$$

$$= \int_0^2 \int_{y=1}^{y=x^2} dy \cdot dx$$

$$= \int_{x=0}^{x=2} [y]_1^{x^2} \cdot dx$$

$$= \int_0^2 (x^2 - 1) \cdot dx$$

$$= \frac{1}{3} [x^3]_0^2 - [x]_0^2$$

$$= \frac{1}{3} (2^3 - 0) - (2 - 0)$$

$$= \frac{2}{3}$$

$$* \int_0^1 \int_0^y dA$$

$$= \int_{y=0}^{y=1} \int_{x=0}^{x=y} dx \cdot dy$$

$$= \int_0^1 [x]_0^y \cdot dy$$

$$= \int_0^1 (y - 0) \cdot dy$$

$$= \frac{1}{2} [y^2]_0^1$$

$$= \frac{1}{2} \times 1^2$$

$$= \frac{1}{2}$$

Iterated Integral:-

Pages:-

Calculate the iterated integral:-

$$(15) \int_1^4 \int_0^2 (6xy - 2x) \cdot dy \cdot dx$$

$$\Rightarrow \int_1^4 \left\{ 6x^2 \left[\frac{1}{2} y^2 \right]_0^2 - 2x \left[y \right]_0^2 \right\} \cdot dx$$

$$\Rightarrow \int_1^4 \{ 3x^2 \times 4 - 2x \times 2 \} \cdot dx$$

$$\Rightarrow 12 \times \frac{1}{3} [x^3]_1^4 - 4 \times \frac{1}{2} [x^2]_1^4$$

$$\Rightarrow 4 \times (4^3 - 1) - 2(4^2 - 1)$$

$$\Rightarrow \underline{222}$$

$$(16) \int_0^1 \int_0^1 (x+y)^2 \cdot dx \cdot dy$$

$$\Rightarrow \int_0^1 \int_0^1 (x^2 + 2xy + y^2) \cdot dx \cdot dy$$

$$= \int_0^1 \left\{ \frac{1}{3} [x^3]_0^1 + 2y \times \frac{1}{2} [x^2]_0^1 + y^2 [x]_0^1 \right\} \cdot dy$$

$$= \int_0^1 \left(\frac{1}{3} + y + y^2 \right) \cdot dy$$

$$= \frac{1}{3} [y]_0^1 + \frac{1}{2} [y^2]_0^1 + \frac{1}{3} [y^3]_0^1$$

$$= \frac{1}{3} + \frac{1}{2} + \frac{1}{3}$$

$$= \underline{\frac{7}{6}}$$

$$(17) \int_0^1 \int_1^2 (x + e^{-y}) \cdot dx dy$$

$$= \int_0^1 \left[\frac{1}{2} x^2 + e^{-y} x \right]_1^2 dy$$

$$= \int_0^1 \left(\frac{3}{2} + e^{-y} \right) dy$$

$$= \left[\frac{3}{2} y + [-e^{-y}] \right]_0^1$$

$$= \frac{3}{2} - (e^{-1} - e^0)$$

$$= \frac{3}{2} + 1 - e^{-1}$$

$$= \frac{5}{2} - \frac{1}{e} = 2.13$$

$$(18) \int_0^{\pi/6} \int_0^{\pi/2} (\sin x + \sin y) dy \cdot dx$$

$$= \int_0^{\pi/6} \left\{ \sin x \left[y \right]_0^{\pi/2} - [\cos y]_0^{\pi/2} \right\} dx$$

$$= \int_0^{\pi/6} \left\{ \frac{\pi}{2} \sin x - [\cos \frac{\pi}{2} - \cos 0] \right\} dx$$

$$= \int_0^{\pi/6} \left\{ \frac{\pi}{2} \sin x + 1 \right\} dx$$

$$= -\frac{\pi}{2} [\cos x]_0^{\pi/6} + [x]_0^{\pi/6}$$

$$= -\frac{\pi}{2} (\cos \frac{\pi}{6} - \cos 0) + \frac{\pi}{6}$$

$$= -\frac{\pi}{2} \times \frac{\sqrt{3}}{2} + \frac{\pi}{2} + \frac{\pi}{6}$$

$$= \frac{2\pi}{3} - \frac{\pi\sqrt{3}}{4} = 0.73$$

$$(19) \int_{-3}^3 \int_0^{\pi/2} (y + y^2 \cos x) dx dy$$

$$= \int_{-3}^3 \left\{ y [x]_0^{\pi/2} + y^2 [\sin x]_0^{\pi/2} \right\} dy$$

$$= \int_{-3}^3 \left\{ \frac{\pi}{2} y + y^2 \sin \frac{\pi}{2} \right\} dy$$

$$= \int_{-3}^3 \frac{\pi}{2} y dy + \int_{-3}^3 y^2 dy$$

$$= 0 + 2 \int_0^3 y^2 dy$$

$$= 2 \times \frac{1}{3} [y^3]_0^3$$

$$= \frac{2}{3} [3^3]$$

$$= 18$$

Let $y = u$
 $f(y) = y$
 $f(-y) = -y$
 $= -f(y)$
 odd function

Let $y = u$
 $f(y) = y^2$
 $f(-y) = (-y)^2$
 $= y^2$
 $= f(y)$
 even function

$$(20) \int_1^3 \int_1^5 \frac{\ln y}{xy} dy dx$$

$$= \int_1^3 \int_1^5 \frac{z}{x} \cdot dz \cdot dx$$

$$= \int_1^3 \frac{1}{x} \times \frac{1}{2} [z^2]_1^5 dx$$

$$= \int_1^3 \left\{ \frac{1}{2} \times (\ln 5)^2 \times \frac{1}{x} \right\} dx$$

$$= \frac{(\ln 5)^2}{2} [\ln x]_1^3$$

Let $z = \ln y$
 $\frac{dz}{dy} = \frac{1}{y}$

$$\frac{1}{y} dy = dz$$

y	1	5
z	0	ln 5

$$= \frac{(\ln 5)^2}{2} \times \ln 3$$

$$= 1.42$$

$$(21) \int_1^4 \int_1^2 \left(\frac{x}{y} + \frac{y}{x} \right) \cdot dy dx$$

$$= \int_1^4 \left\{ x \left[\ln y \right]_1^2 + \frac{1}{x} \times \frac{1}{2} \left[y^2 \right]_1^2 \right\} dx$$

$$= \int_1^4 \left\{ x \ln 2 + \frac{3}{2} \times \frac{1}{x} \right\} \cdot dx$$

$$= \frac{1}{2} \ln 2 \left[x^2 \right]_1^4 + \frac{3}{2} \left[\ln x \right]_1^4$$

$$= \frac{1}{2} \ln 2 (16 - 1) + \frac{3}{2} \ln 4$$

$$= \frac{15}{2} \ln 2 + \frac{3}{2} \ln 4$$

$$= 7.2780$$

Calculate the double
Integral:—

(27) $\iint_R x \sec^2 y \, dA$, $R = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq \pi/4\}$

$$= \int_0^2 \int_0^{\pi/4} n \sec^2 y \, dy \, dx$$

$$= \int_0^2 m [\tan y]_0^{\pi/4} dm$$

$$= \int_0^2 x \, dx$$

$$= \frac{1}{2} [x^2]_0^2$$

$$= \frac{4}{2}$$

$$= 2$$

28) $\iint_R (y + xy^{-2}) \, dA$, $R = \{(x, y) \mid 0 \leq x \leq 2, 1 \leq y \leq 2\}$

$$= \int_0^2 \int_1^2 (y + my^{-2}) dy dx$$

$$= \int_0^2 \frac{1}{2} [y']^2 + m \left[\frac{y^{-2+1}}{-2+1} \right]_0^2 dx$$

$$= \int_0^2 \left(\frac{3}{2} - n \left[\frac{1}{2} - 1 \right] \right) dn$$

$$= \frac{3}{2} [u]_0^2 + \frac{1}{2} \times \frac{1}{2} [u^2]_0^2$$

$$= \frac{2 \times 2}{2} + \frac{1}{4} \times 4 = 4$$

$$(22) \iint_R \frac{xy^2}{x^2+1} dA \quad R = \{(x,y) \mid 0 \leq x \leq 1, -3 \leq y \leq 3\}$$

$$= \int_0^1 \int_{-3}^3 \frac{xy^2}{x^2+1} dy dx = \int_0^1 \int_{-3}^3 \frac{xy^2}{x^2+1} dy dx$$

$$= \int_{-3}^3 \int_0^1 \frac{xy^2}{x^2+1} dx dy = \int_0^1 2 \int_{-3}^3 \frac{xy^2}{x^2+1} dy dx$$

$$= \int_{-3}^3 y^2 \left[\frac{1}{2} \ln(x^2+1) \right]_0^1 dy = \int_0^1 2 \frac{x}{x^2+1} \times \frac{1}{2+1} [y^3]_0^3 dx$$

$$= \int_{-3}^3 \frac{1}{2} y^2 \ln 2 dy = \frac{2 \times 27}{3} \int_0^1 \frac{x}{x^2+1} dx = 18 \times \frac{1}{2} [\ln(x^2+1)]_0^1$$

$$= 2 \int_0^3 \frac{1}{2} \ln 2 y^2 dy = 0 \ln 2$$

$$= 2 \left[\frac{1}{2} \ln 2 \times \frac{1}{3} y^3 \right]_0^3$$

$$= 0 \ln 2$$

$$(23) \iint_R ye^{-xy} dA, \quad R = [0, 2] \times [0, 3]$$

$$= \int_0^2 \int_0^3 ye^{-xy} dy dx$$

$$= \int_0^3 \int_0^2 ye^{-xy} dx dy$$

$$= \int_0^3 y \left[-\frac{1}{y} e^{-xy} \right]_0^2 dy$$

$$= \int_0^3 -[e^{-2y} - e^0] dy$$

$$= [y]_0^3 + \frac{1}{2} [e^{-2y}]_0^3$$

$$= 3 + \frac{1}{2} [e^{-6} - e^0] = \frac{1}{2e^6} + \frac{5}{2} = 2.50$$

Double Integral Over general regions:

Page 81008:

Evaluate the iterated integral:

$$\textcircled{1} \int_1^5 \int_0^x (8x - 2y) dy dx$$

$$\Rightarrow \int_1^5 \left\{ 8x[y]_0^x - 2 \times \frac{1}{2} [y^2]_0^x \right\} dx$$

$$\Rightarrow \int_1^5 (8x^2 - x^2) dx$$

$$\Rightarrow 7 \times \frac{1}{3} [x^3]_1^5$$

$$\Rightarrow \frac{7}{3} [5^3 - 1^3]$$

$$\Rightarrow \frac{868}{3}$$

$$\textcircled{2} \int_0^2 \int_0^{y^2} x^2 y dx dy$$

$$= \int_0^2 \frac{1}{3} y [x^3]_0^{y^2} dy$$

$$= \int_0^2 \frac{1}{3} y^7 dy$$

$$= \frac{1}{3} \times \frac{1}{8} [y^8]_0^2$$

$$= \frac{1}{3} \times \frac{1}{8} \times 2^8$$

$$= \frac{32}{3}$$

$$\textcircled{3} \int_0^2 \int_0^4 x e^{y^3} dx dy$$

$$= \int_0^2 \left[\frac{1}{2} e^{y^3} x^2 \right]_0^4 dy$$

$$= \int_0^2 \frac{1}{2} e^{y^3} \cdot y^2 dy$$

$$= \int_0^2 \frac{1}{2} \times \frac{1}{3} e^z \cdot dz$$

$$= \frac{1}{6} [e^z]_0^2$$

$$= \frac{1}{6} [e^1 - 1]$$

$$= 0.286$$

$$\left. \begin{aligned} \text{Let } z &= y^3 \\ \frac{dz}{dy} &= 3y^2 \\ \frac{1}{3} dz &= y^2 dy \end{aligned} \right\}$$

4	1	0
2	1	0

$$\textcircled{4} \int_0^{\pi/2} \int_0^{\pi} x \sin y dy dx$$

$$= \int_0^{\pi/2} -x [\cos y]_0^{\pi} dx$$

$$= \int_0^{\pi/2} -x [\cos \pi - 1] dx$$

$$= \int_0^{\pi/2} [x - x \cos \pi] dx$$

$$= \frac{1}{2} [x^2]_0^{\pi/2} - \left\{ [x \sin x + \cos x]_0^{\pi/2} \right\}$$

$$= \frac{\pi^2}{8} - \left\{ \frac{\pi}{2} \sin \frac{\pi}{2} - \cos \frac{\pi}{2} + \cos 0 \right\}$$

$$= \frac{\pi^2}{8} - \frac{\pi}{2} + 1$$

$$= 0.662$$

Evaluate the Double Integrals

⑦ $\iint_D \frac{4}{x^2+1} dA, D = \{(x,y) | 0 \leq x \leq 4, 0 \leq y \leq \sqrt{x}\}$

$$= \int_0^4 \int_0^{\sqrt{x}} \frac{4}{x^2+1} dy \cdot dx$$

$$= \int_0^4 \frac{1}{x^2+1} \times \frac{1}{2} [y^2]_0^{\sqrt{x}} dx$$

$$= \int_0^4 \frac{1}{2} \times \frac{1}{x^2+1} x \cdot dx$$

$$= \frac{1}{2} \times \frac{1}{2} [\ln|x^2+1|]_0^4$$

$$= \frac{1}{4} (\ln 17 - \ln 1)$$

$$= 0.7083$$

⑧ $\iint_D (2x+y) dA, D = \{(x,y) | 1 \leq y \leq 2, y-1 \leq x \leq 1\}$

$$= \int_1^2 \int_{y-1}^1 (2x+y) dx dy$$

$$= \int_1^2 \left\{ 2x \frac{1}{2} [x^2]_{y-1}^1 + y [x]_{y-1}^1 \right\} dy$$

$$= \int_1^2 \left\{ (1-(y-1)^2) + y(1-y+1) \right\} dy$$

$$= \int_1^2 \{ 1 - (y^2 - 2y + 1) + 2y - y^2 \} \cdot dy$$

$$= \int_1^2 \{ 1 - y^2 + 2y - 1 + 2y - y^2 \} \cdot dy$$

$$= \int_1^2 (4y - 2y^2) \cdot dy$$

$$= 4 \times \frac{1}{2} [y^2]_1^2 - 2 \times \frac{1}{3} [y^3]_1^2$$

$$= 2(4-1) - \frac{2}{3}(8-1)$$

$$= \frac{4}{3}$$

17) $\iint_D x \cos y \, dA$. D is bounded by $y=0, y=\pi^2, x=1$

Find the limit as

$$\pi^2 \rightarrow 0$$

$$\pi = 0$$

So,

$$= \int_0^1 \int_0^{\pi^2} x \cos y \cdot dy \, dx$$

$$= \int_0^1 x [\sin y]_0^{\pi^2} dx$$

$$= \int_0^1 x \sin \pi^2 \cdot dx$$

$$= \int_0^1 \sin z \times \frac{1}{2} dz$$

$$= -\frac{1}{2} [\cos z]_0^1$$

$$= -\frac{1}{2} \cos 1 + \frac{1}{2} = 7.62 \times 10^{-5}$$

Let

$$z = \pi^2$$

$$\frac{dz}{dx} = 2\pi$$

$$\frac{1}{2} dz = \pi \cdot dx$$

x	1	0
z	1	0

18) $\iint_D (x^2 + 2y) \, dA$, D is bounded by $y=x$, $y=x^3$, $x>0$

$\Rightarrow$ Find the limit of x —

$$x^3 = x$$

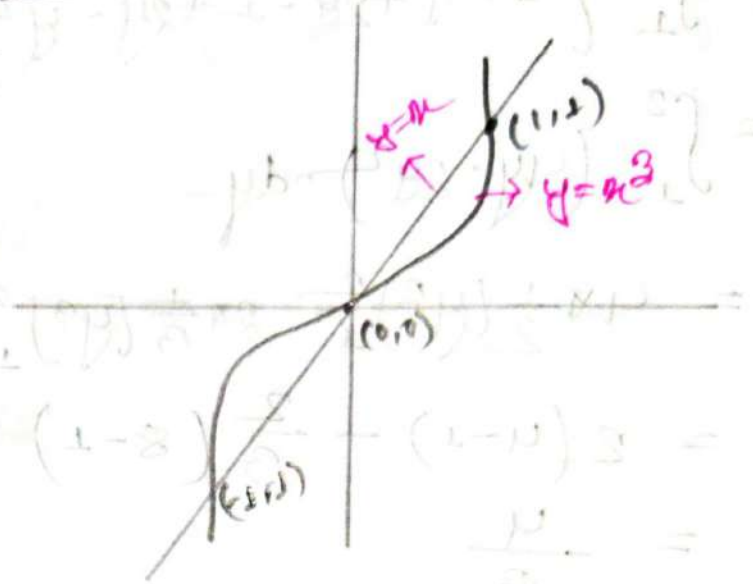
$$x^3 - x = 0$$

$$x(x^2 - 1) = 0$$

$$\therefore x = 0 \text{ or } x = \pm 1$$

So, the limit is —

$$x = 0, x = 1$$



now $\int_0^1 \int_x^{x^3} (x^2 + 2y) \, dy \, dx$

$$= \int_0^1 \left\{ x^2 y \Big|_x^{x^3} + 2 \times \frac{1}{2} [y^2]_x^{x^3} \right\} dx$$

$$= \int_0^1 \left\{ x^2 (x^3 - x) + [x^6 - x^2] \right\} dx$$

$$= \int_0^1 (x^5 - x^3 + x^6 - x^2) dx$$

$$= \left[\frac{1}{6} x^6 - \frac{1}{4} x^4 + \frac{1}{7} x^7 - \frac{1}{3} x^3 \right]_0^1$$

$$= \frac{1}{6} - \frac{1}{4} + \frac{1}{7} - \frac{1}{3}$$

$$= -\frac{23}{84}$$

$$= \int_0^1 \int_{x^3}^x (x^2 + 2y) \, dy \, dx$$

$$= \int_0^1 \left\{ x^2 y \Big|_{x^3}^x + [y^2]_{x^3}^x \right\} dx$$

$$= \int_0^1 \left\{ x^3 - x^5 + x^2 - x^6 \right\} dx$$

$$= \left[\frac{1}{4} x^4 - \frac{1}{6} x^6 + \frac{1}{3} x^3 - \frac{1}{7} x^7 \right]_0^1$$

$$= \frac{1}{4} - \frac{1}{6} + \frac{1}{3} - \frac{1}{7}$$

$$= \frac{23}{84}$$

Triple Integrals

Q Evaluate the iterated integrals—

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$$\textcircled{3} \int_0^2 \int_0^{z^2} \int_0^{y-z} (2x-y) \, dx \, dy \, dz$$

$$= \int_0^2 \int_0^{z^2} \int_0^{y-z} (2x-y) \, dx \, dy \, dz$$

$$= \int_0^2 \int_0^{z^2} \left\{ 2x \cdot \frac{1}{2} [x^2]_0^{y-z} - y [x]_0^{y-z} \right\} dy \, dz$$

$$= \int_0^2 \int_0^{z^2} \left\{ (y-z)^2 - y^2 + yz \right\} dy \, dz$$

$$= \int_0^2 \int_0^{z^2} \left\{ y^2 - 2yz + z^2 - y^2 + yz \right\} dy \, dz$$

$$= \int_0^2 \int_0^{z^2} (z^2 - yz) dy \, dz$$

$$= \int_0^2 \left\{ z^2 [y]_0^{z^2} - z \cdot \frac{1}{2} [y^2]_0^{z^2} \right\} dz$$

$$= \int_0^2 \left(z^4 - \frac{1}{2} z^5 \right) dz$$

$$= \frac{1}{5} [z^5]_0^2 - \frac{1}{2} \times \frac{1}{6} [z^6]_0^2$$

$$= \frac{1}{5} \times 2^5 - \frac{1}{2} \times \frac{1}{6} \times 2^6$$

$$= \frac{16}{15}$$

$$(4) \int_0^1 \int_4^{24} \int_0^{n+4} 6xy \, dv$$

$$= \int_0^1 \int_4^{24} \int_0^{n+4} 6xy \, dz \, dn \, dy$$

$$= \int_0^1 \int_4^{24} 6xy [z]_0^{n+4} \, dn \, dy$$

$$= \int_0^1 \int_4^{24} (6x^2y + 6xy^2) \, dn \, dy$$

$$= \int_0^1 \left\{ 6xy \left[\frac{n^3}{3} \right]_4^{24} + 6y^2 \left[\frac{1}{2} n^2 \right]_4^{24} \right\} dy$$

$$= \int_0^1 \left\{ 24(8y^3 - y^3) + 3y^2(48 - 4^2) \right\} dy$$

$$= \int_0^1 (14y^4 + 2y^4) \, dy$$

$$= 23 \times \frac{1}{5} [y^5]_0^1$$

$$= \frac{23}{5}$$

$$(5) \int_1^2 \int_0^{2z} \int_0^{enx} ne^{-y} \, dy \, dn \, dz$$

$$= \int_1^2 \int_0^{2z} -n[e^{-y}]_0^{enx} \, dn \, dz$$

$$= \int_1^2 \int_0^{2z} \{-ne^{-enx} + n\} \, dn \, dz$$

$$= \int_1^2 \left[\frac{1}{2} [n^2]_0^{2z} - \int_0^{2z} n \times \frac{1}{n} \, dn \right] dz$$

$$\begin{aligned} &= \int e^{\ln(n)} \cdot dn \\ &= \int e^{\ln(n^{\frac{1}{n}})} \cdot dn \\ &= \int n^{\frac{1}{n}} \cdot dn \end{aligned}$$

$$= \int_1^2 \left\{ \frac{1}{2} 4z^2 - [x]_0^{2z} \right\} dz$$

$$= \int_1^2 (2z^2 - 2z) dz$$

$$= 2 \times \frac{1}{3} [z^3]_1^2 - 2 \times \frac{1}{2} [z^2]_1^2$$

$$= \frac{2}{3} (2^3 - 1) - (4 - 1)$$

$$= \frac{5}{3}$$

$$\textcircled{6} \int_0^1 \int_0^1 \int_0^{\sqrt{1-z^2}} \frac{z}{y+1} dx \cdot dz \cdot dy$$

$$= \int_0^1 \int_0^1 \frac{z}{y+1} [x]_0^{\sqrt{1-z^2}} dz \cdot dy$$

$$= \int_0^1 \int_0^1 \frac{z}{y+1} \times (\sqrt{1-z^2}) dz \cdot dy$$

$$= \int_0^1 \int_1^0 -\frac{1}{2} \times \frac{1}{y+1} \sqrt{a} \cdot da \cdot dy$$

$$= \int_0^1 -\frac{1}{2} \times \frac{1}{y+1} \left[\frac{a^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_1^0 dy$$

$$= -\frac{1}{2} \int_0^1 \frac{1}{y+1} \times \frac{2}{3} [a^{\frac{3}{2}}]_1^0 dy$$

$$= -\frac{1}{3} \int_0^1 \frac{1}{y+1} [0^{\frac{3}{2}} - 1^{\frac{3}{2}}] dy$$

Let $a = 1 - z^2$
 $\frac{da}{dz} = -2z$
 $-\frac{1}{2} da = z \cdot dz$

a	1	0
z	0	1

$$= -\frac{1}{3} \Big|_0^1 - \left(\frac{1}{y+1} \right) dy$$

$$= \frac{1}{3} [\ln|y+1|]_0^1$$

$$= \frac{1}{3} \ln 2$$

$$= 0.2310$$

$$(7) \int_0^\pi \int_0^1 \int_0^{\sqrt{1-z^2}} z \sin x \, dy \, dz \, dx$$

$$= \int_0^\pi \int_0^1 z \sin x \left[y \right]_0^{\sqrt{1-z^2}} dz \, dx$$

$$= \int_0^\pi \int_0^1 z \sin x \sqrt{1-z^2} \, dz \, dx$$

$$= \int_0^\pi \int_1^0 -\frac{1}{2} \sin x \sqrt{a} \cdot da \, dx$$

$$= -\frac{1}{2} \int_0^\pi \sin x \left[\frac{a^{\frac{1}{2}+1}}{\frac{1}{2}+1} \right]_1^0 dx$$

$$= -\frac{1}{2} \int_0^\pi \left\{ \frac{2}{3} \sin x \left[0^{\frac{3}{2}} - \frac{3}{2} \right] \right\} \cdot dx$$

$$= -\frac{1}{3} \int_0^\pi -\sin x \, dx$$

$$= -\frac{1}{3} [\cos x]_0^\pi$$

$$= -\frac{1}{3} \cos \pi + \frac{1}{3}$$

$$= \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

Let

$$a = 1 - z^2$$

$$\frac{da}{dz} = -2z$$

$$-\frac{1}{2} da = z \cdot dz$$

z	0	1
a	1	0

#Double Integrals Over General Regions:

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Example 18- Evaluate $\iint_D (x+2y)$, where D is the region bounded by the parabolas $y=2x^2$ and $y=1+x^2$

⇒ Find the intersection points:

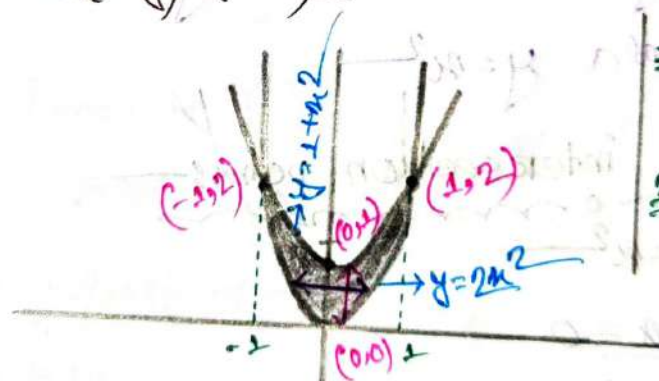
$$2x^2 = 1 + x^2$$

$$x^2 = 1$$

$$x = \pm 1$$

$$x = 1, y = 2; (x, y) = (1, 2)$$

$$x = -1, y = 2; (x, y) = (-1, 2)$$



1. Vertical Strip
y change

2. Horizontal Strip
x change.

$$\text{So, } \iint_D (x+2y) dy \cdot dx$$

$$= \int_{-1}^1 x \left[y \right]_{2x^2}^{1+x^2} + 2x \left[\frac{y^2}{2} \right]_{2x^2}^{1+x^2} \cdot dx$$

$$= \int_{-1}^1 x \left\{ 1+x^2-2x^2 \right\} + (1+x^2)^2 - (2x^2)^2 \cdot dx$$

$$= \int_{-1}^1 (x+x^3-2x^3+1+2x^2+x^4-4x^4) \cdot dx$$

$$= \int_{-1}^1 (x - x^3 + 1 + 2x^2 - 3x^4) \cdot dx$$

$$= 2 \int_0^1 (1 + 2x^2 - 3x^4) \cdot dx$$

$$= 2 \left[x + \frac{2}{3} x^3 - \frac{3}{5} x^5 \right]_0^1$$

$$= 2 \left(1 + \frac{2}{3} - \frac{3}{5} \right)$$

$$= \frac{32}{15}$$

Example: 2 Find the Volume of the solid region that under the paraboloid $z = x^2 + y^2$ and above the region D in the xy plane bounded by the line $y = 2x$ and the parabola $y = x^2$

⇒ Find the intersection points:

$$2x = x^2$$

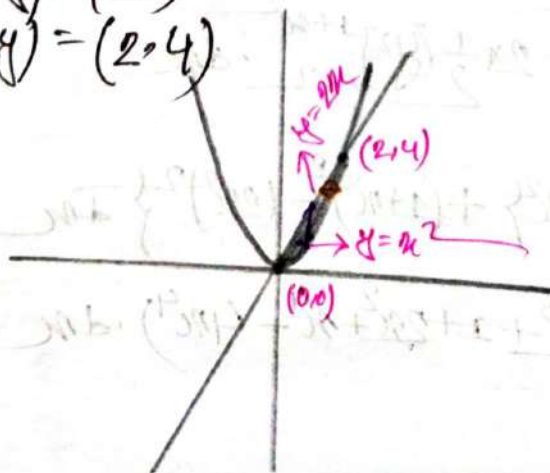
$$x^2 - 2x = 0$$

$$x(x - 2) = 0$$

$$\therefore x = 0, 2$$

$$x = 0, y = 0; (x, y) = (0, 0)$$

$$x = 2, y = 4; (x, y) = (2, 4)$$



50. $\int_0^2 \int_{x^2}^{2x} (x^2 + y^3) \cdot dy \, dx$

$$= \int_0^2 \left\{ x^2 [y]_{x^2}^{2x} + \frac{1}{3} [y^3]_{x^2}^{2x} \right\} dx$$

$$= \int_0^2 \left(2x^3 - x^4 + \frac{1}{3} 8x^3 - \frac{1}{3} x^6 \right) dx$$

$$= \int_0^2 \left(\frac{14}{3} x^3 - x^4 - \frac{1}{3} x^6 \right) dx$$

$$= \left[\frac{14}{3} \times \frac{1}{4} x^4 - \frac{1}{5} x^5 - \frac{1}{3} \times \frac{1}{7} x^7 \right]_0^2$$

$$= \frac{14}{12} \times 2^4 - \frac{1}{5} 2^5 - \frac{1}{21} 2^7$$

$$= \frac{216}{35}$$

Example 3: — Evaluate $\iint_D xy \, dA$ where D is the region bounded by the line $y = x - 1$ and the parabola $y^2 = 2x + 6$

⇒ Rewrite the equations —

$$x = y + 1$$

$$x = \frac{1}{2} y^2 - 3$$

Find the intersection point:

$$1 + y = \frac{1}{2} y^2 - 3$$

$$y + 1 = \frac{y^2 - 6}{2}$$

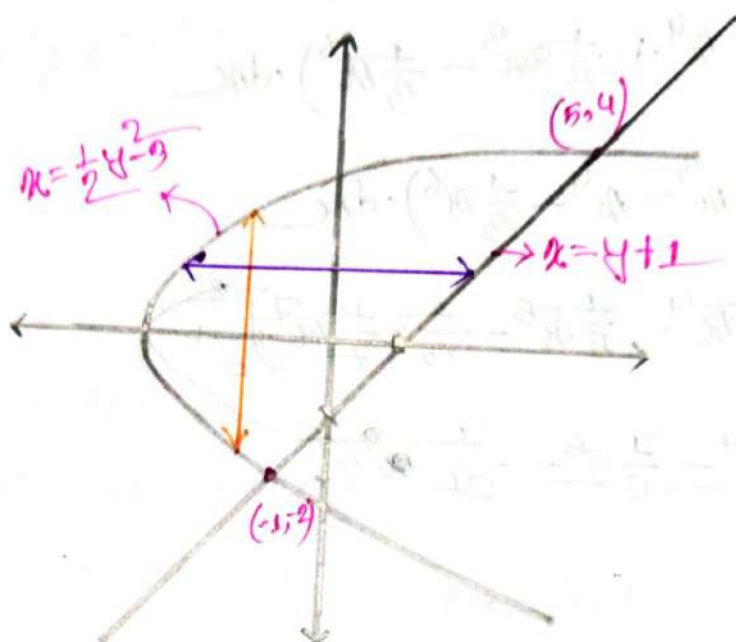
$$2y + 2 = y^2 - 6$$

$$y^2 - 2y - 8 = 0$$

$$y = -2, 4$$

$$y = -2, x = -1; (x, y) = (-1, -2)$$

$$y = 4, x = 5; (x, y) = (5, 4)$$



$$\underline{\text{So,}} = \int_{-2}^4 \int_{\frac{1}{2}y^2-3}^{y+1} xy \, dx \, dy$$

$$= \int_{-2}^4 \frac{1}{2} y \left[x^2 y + 1 \right]_{\frac{1}{2}y^2-3}^{\frac{1}{2}y^2-3} dy$$

$$= \int_{-2}^4 \frac{1}{2} y \left\{ (y+1)^2 - \left(\frac{1}{2}y^2-3 \right)^2 \right\} dy$$

$$= \frac{1}{2} \int_{-2}^4 y \left(y^2 + 2y + 1 - \frac{1}{4}y^4 + 3y^2 - 9 \right) dy$$

$$= \frac{1}{2} \int_{-2}^4 \left(4y^3 + 2y^2 - 2y - \frac{1}{4}y^5 \right) dy$$

$$= \frac{1}{2} \left[4 \times \frac{1}{4} y^4 + \frac{2}{3} y^3 - 4y^2 - \frac{1}{24} y^6 \right]_{-2}^4$$

$$= \frac{1}{2} \left[(4^4 - 2^4) + \frac{2}{3} (4^3 - 2^3) - 4 \times (4^2 - 2^2) - \frac{1}{24} (4^6 - 2^6) \right]$$

$$= \frac{1}{2} \left(240 + \frac{2}{3} \times 72 - 4 \times 12 - \frac{1}{24} \times 4032 \right)$$

$$= \frac{1}{2} (240 + 48 - 48 - 168)$$

$$= \frac{1}{2} \times 72$$

$$= 36$$

Double integral in polar
Coordinates: _____

We know: _____

$$\left. \begin{aligned} x &= r \cos \theta \\ \frac{dx}{d\theta} &= \frac{d}{d\theta} r \cos \theta \\ &= -r \sin \theta \end{aligned} \right\} \begin{aligned} y &= r \sin \theta \\ \frac{dy}{d\theta} &= \frac{d}{d\theta} r \sin \theta \\ &= r \cos \theta \end{aligned}$$

$$\boxed{dx dy = r dr d\theta}$$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

$$= \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix}$$

$$= r \cos^2 \theta + r \sin^2 \theta$$

$$= r (\cos^2 \theta + \sin^2 \theta)$$

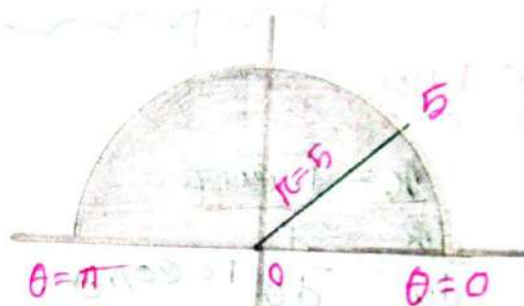
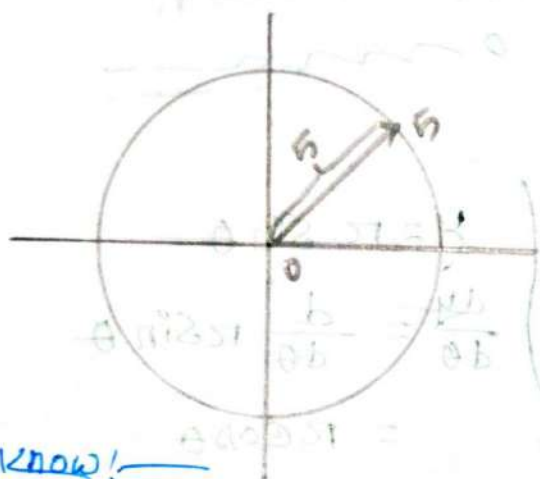
$$= r$$

$$\therefore dx dy = r dr d\theta$$

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Evaluate the given integral by changing to polar coordinates.

7. $\iint_D x^2 y \, dA$, where D is the top half of the disk with center the origin and radius 5



We know!

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dA = dx dy = r dr d\theta$$

So
$$= \int_0^\pi \int_0^5 r^2 \cos^2 \theta \times r \sin \theta \cdot r dr d\theta$$

$$= \int_0^\pi \int_0^5 r^4 \cos^2 \theta \cdot \sin \theta \cdot dr d\theta$$

$$= \int_0^\pi \cos^2 \theta \cdot \sin \theta \times \frac{1}{5} [r^5]_0^5 d\theta$$

$$= -625 \times \frac{1}{3} [\cos^3 \theta]_0^\pi$$

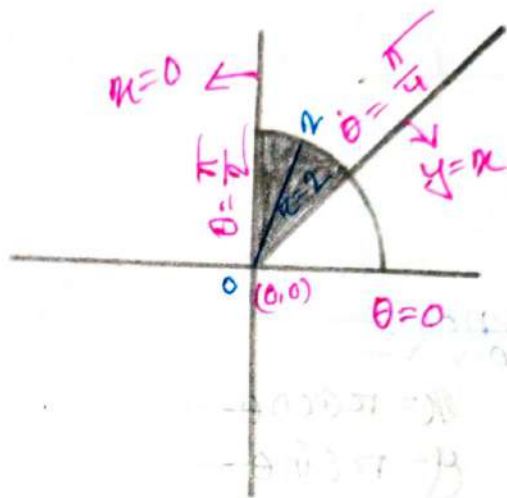
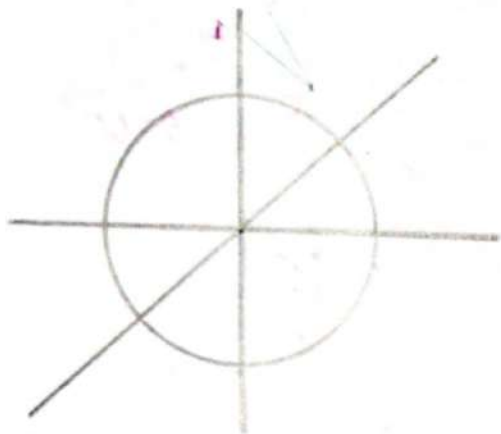
$$= -\frac{625}{3} [\cos(\pi)^3 - \cos(0)^3]$$

$$= 416.667$$

Let

$$b = \cos \theta$$
$$\frac{db}{d\theta} = -\sin \theta$$
$$db = \sin \theta \cdot d\theta$$

⑧ $\iint_R (2x-y) dA$, where R is the region in the first quadrant enclosed by the circle $x^2+y^2=4$, and the line $y=x$ and $x=0$



We know—

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dA = dx dy = r dr d\theta$$

$$\text{So, } = \int_{\pi/4}^{\pi/2} \int_0^2 (2r \cos \theta - r \sin \theta) r dr d\theta$$

$$= \int_{\pi/4}^{\pi/2} \int_0^2 r^2 (2 \cos \theta - \sin \theta) dr d\theta$$

$$= \int_{\pi/4}^{\pi/2} \frac{8}{3} (2 \cos \theta - \sin \theta) d\theta$$

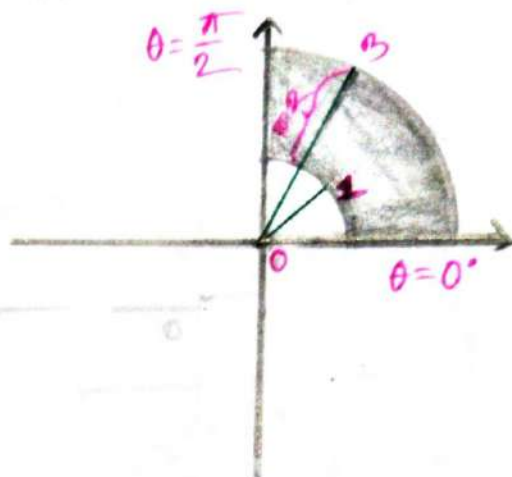
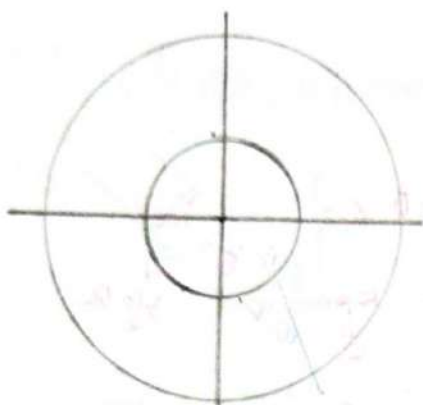
$$= \frac{16}{3} [\sin \theta]_{\pi/4}^{\pi/2} + \frac{8}{3} [\cos \theta]_{\pi/4}^{\pi/2}$$

$$= \frac{16}{3} \left[\sin \frac{\pi}{2} - \sin \frac{\pi}{4} \right] + \frac{8}{3} \left[\cos \frac{\pi}{2} - \cos \frac{\pi}{4} \right]$$

$$= \frac{16}{3} \left(1 - \frac{1}{\sqrt{2}} \right) + \frac{8}{3} \left(0 - \frac{1}{\sqrt{2}} \right)$$

$$= -0.13235$$

② $\iint \sin(x^2+y^2) dA$, where $\mathcal{R}$ is the region in the first quadrant between the circles with center the origin and radii 1 and 3



we know:-

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dA = dx dy = r \cdot dr d\theta$$

$$x^2 + y^2 = r^2$$

$$\underline{\text{So,}} = \int_0^{\pi/2} \int_1^3 \sin r^2 \cdot r \, dr d\theta$$

$$= \int_0^{\pi/2} -\frac{1}{2} \left[\cos r^2 \right]_1^3 d\theta$$

$$= \int_0^{\pi/2} -\frac{1}{2} [\cos 9 - \cos 1] d\theta$$

$$= -\frac{1}{2} [\cos 9 - \cos 1] \times \left[\theta \right]_0^{\pi/2}$$

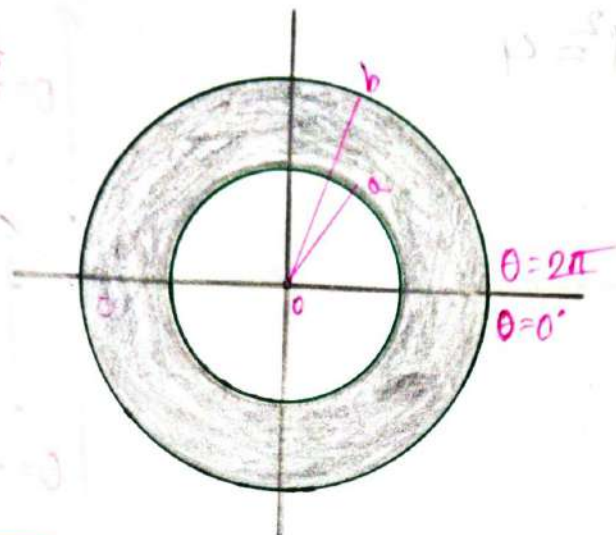
$$= -\frac{1}{2} [\cos 9 - \cos 1] \times \frac{\pi}{2}$$

$$= \frac{\pi}{4} (\cos 1 - \cos 9)$$

$$= 1.1309$$

$$\left\{ \begin{array}{l} \underline{\text{Let}} \\ z = r^2 \\ \frac{dz}{dr} = 2r \\ \frac{1}{2} dz = r \cdot dr \end{array} \right.$$

(20) $\iint_R \frac{y^2}{x^2+y^2} dA$, where R is the region that lies between the circles $x^2+y^2=a^2$ and $x^2+y^2=b^2$ with $0 < a < b$



We know!

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$x^2 + y^2 = r^2$$

$$dA = dx dy = r dr d\theta$$

So, $\int_0^{2\pi} \int_a^b \frac{r^2 \sin^2 \theta}{r^2} r dr d\theta$

$$= \int_0^{2\pi} \frac{1}{2} \sin^2 \theta [r^2]_a^b d\theta$$

$$= \frac{1}{2} (b^2 - a^2) \int_0^{2\pi} \frac{1}{2} (1 - \cos 2\theta) d\theta$$

$$= \frac{1}{4} (b^2 - a^2) [\theta]_0^{2\pi} - \frac{1}{4} \times \frac{1}{2} (b^2 - a^2) [\sin 2\theta]_0^{2\pi}$$

$$= \frac{1}{4} (b^2 - a^2) \times 2\pi - \frac{1}{8} (b^2 - a^2) (\sin 4\pi - \sin 0)$$

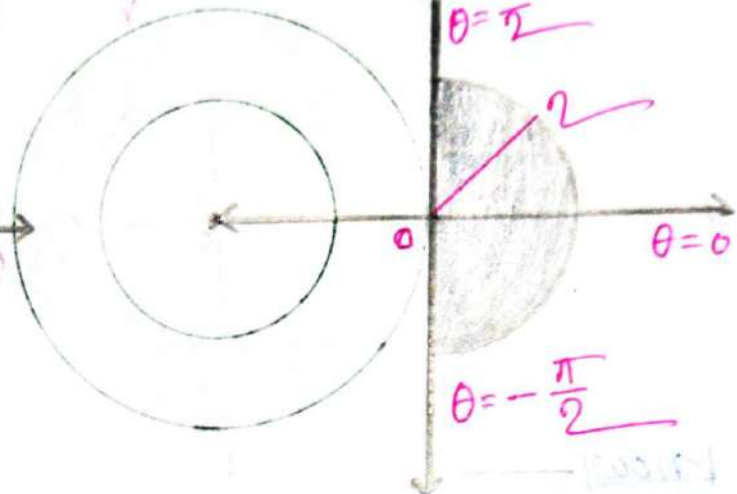
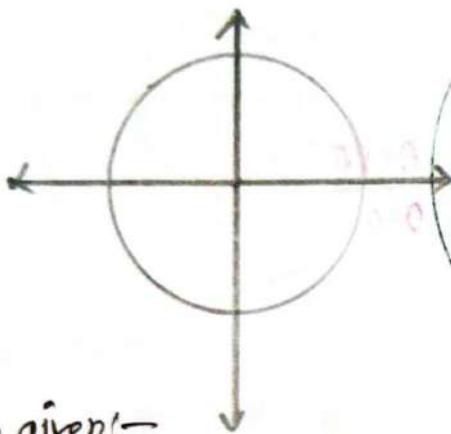
$$= \frac{1}{4} (b^2 - a^2) 2\pi - 0$$

$$= \frac{\pi}{2} (b^2 - a^2)$$

21) $\iint_D e^{-x^2-y^2} \cdot dA$ where D is the region bounded by the semi-circle $x = \sqrt{4-y^2}$ and the y -axis/
 $x=0$

$\Rightarrow$ re write the equation:-

$$x^2 + y^2 = 4$$



In given:-

$$x = \sqrt{4-y^2}$$

$$x = -(2-y) / x = (2-y)$$

We know:-

$$x = r \cos \theta \quad \int x^2 + y^2 = r^2$$

$$y = r \sin \theta$$

$$dA = dx dy = r \cdot dr d\theta$$

$$\underline{\text{So,}} = \int_{-\pi/2}^{\pi/2} \int_0^2 e^{-(x^2+y^2)} \cdot dA$$

$$= \int_{-\pi/2}^{\pi/2} \int_0^2 e^{-r^2} \cdot r \cdot dr d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \frac{1}{2} [e^{-r^2}]_0^2 d\theta$$

$$\left\{ \begin{array}{l} \underline{\text{let}} \\ z = r^2 \\ \frac{dz}{dr} = 2r \\ \frac{1}{2} dz = r \cdot dr \end{array} \right.$$

$$= -\frac{1}{2} \int_{-\pi/2}^{\pi/2} (e^{-4} - e^0) d\theta$$

$$= -\frac{1}{2} (e^{-4} - e^0) \times [\theta]_{-\pi/2}^{\pi/2}$$

$$= -\frac{1}{2} (e^{-4} - 1) \times \left(\frac{\pi}{2} - \left(-\frac{\pi}{2} \right) \right)$$

$$= \frac{\pi}{2} (1 - e^{-4})$$

$$= 1.5420$$

Change of variable in a Double
Integral: —

Pages 1060: —

Q3) $\iint_R \frac{x-2y}{3x-y} \cdot dA$ where R is the parallelogram enclosed by the lines $x-2y=0$, $x-2y=4$, $3x-y=1$ and $3x-y=8$

$\Rightarrow$ Let, $u = x-2y$; $u=0, 4$

$v = 3x-y$; $v=1, 8$

We know: —

Jacobian matrix $|J|$

$$|J| = \frac{du dv}{dx dy} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -2 \\ 3 & -1 \end{vmatrix}$$

$$= -1 + 6$$

$$= -2 + 7 = 5$$

$$\therefore \frac{dudv}{dxdy} = -2 + 7 = 5$$

$$dxdy =$$

$$\frac{1}{5} dudv$$

So we know / Is given:-

$$= \iint_R \frac{x-2y}{3x+y} \cdot dA$$

$$= \int_0^4 \int_{\frac{1}{5}u}^8 \frac{u}{v} dv du$$

$$= \int_0^4 \left[\frac{u}{v} \ln v \right]_{\frac{1}{5}u}^8 du$$

$$= \int_0^4 \frac{1}{5} u \ln 8 du$$

$$= \frac{1}{2} \ln 8 [u^2]_0^4 \times \frac{1}{5}$$

$$= \frac{1}{2} \times 16 \ln 8 \times \frac{1}{5}$$

$$= 8 \ln 8 \times \frac{1}{5}$$

$$= \frac{8}{5} \ln 8$$

(24) $\iint_R (x+y) e^{x^2-y^2} \cdot dA$ where R is the rectangle enclosed by the lines $x-y=0$, $x-y=4/2$, $x+y=0$, $x+y=3$

Let's
mon

$$u = x - y \quad u = (0, 2)$$

$$v = x + y \quad v = (0, 3)$$

We know:-
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$$|J| = \frac{du dv}{dx dy} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix}$$

$$= 1 + 1$$

$$= 2$$

$$\therefore dx dy = \frac{1}{2} du dv$$

We know/ Is given:-
mon

$$= \iint_D (x+y) e^{x^2-y^2} \cdot dA$$

$$= \int_0^2 \int_0^3 v e^{uv} \frac{1}{2} dv \cdot du$$

$$= \int_0^2 \int_0^3 \frac{1}{2} v e^{uv} dv \cdot du$$

$$= \int_0^3 \int_0^2 \frac{1}{2} v e^{uv} du dv$$

$$= \int_0^3 \frac{1}{2} \left[v \times \frac{1}{v} e^{uv} \right]_0^2 dv$$

$$= \frac{1}{2} \int_0^3 (e^{2v} - e^0) \cdot dv$$

$$= \frac{1}{2} \times \left\{ \left[\frac{1}{2} e^{2v} - 1 \right]_0^3 \right\}$$

$$= \frac{1}{2} \times \frac{1}{2} (e^6 - e^0) - \frac{1}{2} \times 3$$

$$= \frac{1}{4} (e^6 - 1) - \frac{3}{2}$$

$$= \frac{1}{4} e^6 - \frac{7}{4}$$

$$= 99.107$$

#Application!

Ch 25

Application of Double Integrals mass and Center of mass

Formula:—

$$\text{mass } m = \iint_D \delta(x, y) dA$$

Center of mass $(\bar{x}, \bar{y})$:—

$$\bar{x} = \frac{1}{m} \iint_D x \delta(x, y) dA$$

$$\bar{y} = \frac{1}{m} \iint_D y \delta(x, y) dA$$

Where, $\delta(x, y) dA \rightarrow \text{density}$.

Pages 10218
mon

[B-8] — Find the mass and center of mass of the lamina that occupies the region D and has the given density function δ .

$$\textcircled{B} \quad D = \{(x, y) \mid 1 \leq x \leq 3, 1 \leq y \leq 4\}; \delta(x, y) = ky^2$$

we know:—

$$\text{mass, } m = \iint_D \delta(x, y) dA$$

$$= \int_1^3 \int_1^4 ky^2 \cdot dy dx$$

$$= k \int_1^3 \left[\frac{y^3}{3} \right]_1^4 dx$$

$$= \frac{63k}{3} \left[x \right]_1^3 = 42k$$

We know:-
mom

center of mass. $(\bar{x}, \bar{y})$

$$\bar{x} = \frac{1}{m} \iint_D x \delta(x, y) dA$$

$$= \frac{1}{42k} \int_1^3 \int_1^{3+k} kxy^2 dy dx$$

$$= \frac{k}{42k} \int_1^3 x \left[\frac{y^3}{3} \right]_1^{3+k} dx$$

$$= \frac{1}{42} \int_1^3 x \frac{1}{3} (9^3 - 1) dx$$

$$= \frac{21}{42} \left[\frac{1}{2} x^2 \right]_1^3$$

$$= \frac{21 \times 8}{42 \times 2}$$

$$= 2$$

$$\bar{y} = \frac{1}{m} \iint_D y \delta(x, y) dA$$

$$= \frac{1}{42k} \int_1^3 \int_1^{3+k} y \cdot kxy^2 dy dx$$

$$= \frac{k}{42k} \int_1^3 \left[\frac{xy^4}{4} \right]_1^{3+k} dx$$

$$= \frac{255}{42 \times 4} [x]_1^3$$

$$= \frac{255 \times 2}{42 \times 4}$$

$$= \frac{85}{28}$$

$$\therefore m = 42k$$

$$\text{Center of mass } (\bar{x}, \bar{y}) = \left(2, \frac{85}{28} \right)$$

④ $D = \{(x, y) \mid 0 \leq x \leq a, 0 \leq y \leq b\}; \rho(x, y) = 1 + x^2 + y^2$

We know—

$$\text{mass } m = \iint_D \rho(x, y) \cdot dA$$

$$= \int_0^a \int_0^b (1 + x^2 + y^2) \cdot dy \, dx$$

$$= \int_0^a \left[y + x^2 y + \frac{1}{2} y^2 \right]_0^b dx$$

$$= \int_0^a \left[b + x^2 b + \frac{1}{2} b^2 \right]_0^b dx$$

$$= \left[bx + \frac{1}{2+1} x^3 b + \frac{1}{2} b^2 x \right]_0^a$$

$$= ab + \frac{1}{3} a^3 b + \frac{1}{2} ab^2$$

$$= \frac{1}{6} ab(3 + a^2 + b^2)$$

Center of mass $(\bar{x}, \bar{y})$

$$\therefore \bar{x} = \frac{1}{m} \iint_D x \rho(x, y) \cdot dA$$

$$= \frac{3}{ab(3 + a^2 + b^2)} \int_0^a \int_0^b (x + x^3 + xy^2) dy \, dx$$

$$= \frac{3}{ab(3 + a^2 + b^2)} \int_0^a \left[xy \right]_0^b + \left[x^3 y + x \cdot \frac{1}{3} y^3 \right]_0^b dx$$

$$= \frac{3}{ab(3 + a^2 + b^2)} \int_0^a \left(bx + x^3 b + \frac{1}{3} b^3 x \right) dx$$

$$= \frac{3}{ab(3 + a^2 + b^2)} \left[\frac{1}{2} bx^2 + \frac{1}{4} x^4 b + \frac{1}{3} \cdot \frac{1}{2} b^3 x^2 \right]_0^a$$

$$= \frac{3}{ab(3+a^2+b^2)} \left(\frac{1}{2} a^2 b + \frac{1}{4} a^4 b + \frac{1}{6} a^2 b^3 \right)$$

$$= \frac{3}{ab(3+a^2+b^2)} \times \frac{1}{12} a^2 b (6 + 3a^2 + 2b^2)$$

$$= \frac{3a^2 b (6 + 3a^2 + 2b^2)}{ab \cdot 12 (3 + a^2 + b^2)}$$

$$= \frac{a(6 + 3a^2 + 2b^2)}{4(3 + a^2 + b^2)}$$

$$\bar{y} = \frac{1}{m} \iint_D y \rho(x, y) \, dA$$

$$= \frac{3}{ab(3+a^2+b^2)} \int_0^a \int_0^b (y + x^2 y + y^3) \, dy \, dx$$

$$= \frac{3}{ab(3+a^2+b^2)} \int_0^a \left[\frac{1}{2} y^2 + \frac{1}{2} x^2 y^2 + \frac{1}{4} y^4 \right]_0^b \, dx$$

$$= \frac{3}{ab(3+a^2+b^2)} \left[\frac{1}{2} b^2 x + \frac{1}{6} b^2 x^3 + \frac{1}{4} b^4 x \right]_0^a$$

$$= \frac{3}{ab(3+a^2+b^2)} \times \left[\frac{1}{2} ab^2 + \frac{1}{6} a^3 b^2 + \frac{1}{4} ab^4 \right]$$

$$= \frac{3}{ab(3+a^2+b^2)} \times \frac{1}{12} ab^2 (6 + 2a^2 + 3b^2)$$

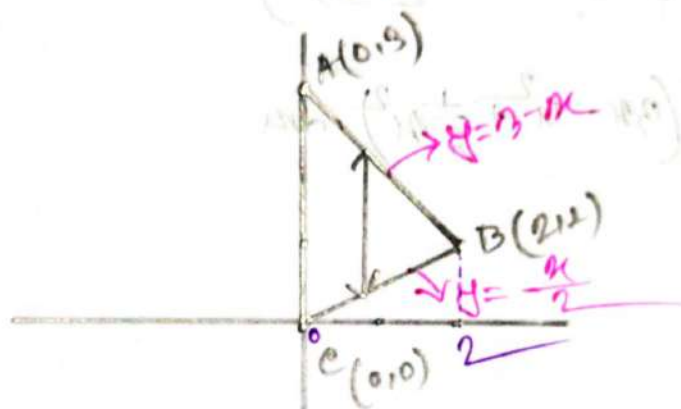
$$= \frac{3ab^2(6 + 2a^2 + 3b^2)}{12ab(3 + a^2 + b^2)}$$

$$= \frac{b(6 + 2a^2 + 3b^2)}{4(3 + a^2 + b^2)}$$

So, mass $m = \frac{1}{3} ab(3+a^2+b^2)$

center of mass $(\bar{x}, \bar{y}) = \left(\frac{a(b+3a^2+2b^2)}{4(3+a^2+b^2)}, \frac{b(b+2a^2+3b^2)}{4(3+a^2+b^2)} \right)$

(5) D is the triangular region with vertices $(0,0)$, $(2,1)$, $(0,3)$; $\delta(x,y) = x+y$



Equation of points A and B:

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2}$$

$$\Rightarrow \frac{x-0}{0-2} = \frac{y-3}{3-1}$$

$$\Rightarrow 2x = -2y + 6$$

$$\Rightarrow x+y = 3$$

Equation of points B and C:

$$\frac{x-x_1}{x_1-x_2} = \frac{y-y_1}{y_1-y_2}$$

$$\Rightarrow \frac{x-2}{2-0} = \frac{y-1}{1-0}$$

$$\Rightarrow x-2 = 2y-2$$

$$\Rightarrow x-2y = 0$$

we know:

$$m = \iint_D \rho(x, y) dA$$

$$= \int_0^2 \int_{\frac{x}{2}}^{3-x} (x+y) dy dx$$

$$= \int_0^2 \left[xy + \frac{1}{2}y^2 \right]_{\frac{x}{2}}^{3-x} dx$$

$$= \int_0^2 \left\{ x(3-x-\frac{x}{2}) + \frac{1}{2} \left[(3-x)^2 - \left(\frac{x}{2}\right)^2 \right] \right\} dx$$

$$= \int_0^2 \left\{ (3x - x^2 - \frac{x^2}{2}) + \frac{1}{2} (9 - 6x + x^2 - \frac{x^2}{4}) \right\} dx$$

$$= \int_0^2 \left(3x - x^2 - \frac{x^2}{2} + \frac{1}{2} \cdot 9 - 3x + \frac{1}{2}x^2 - \frac{1}{8}x^2 \right) dx$$

$$= \int_0^2 \left(\frac{9}{2} - \frac{3x^2}{8} \right) dx$$

$$= \left[\frac{9}{2}x - \frac{3}{8} \cdot \frac{1}{3}x^3 \right]_0^2$$

$$= \left[\frac{9}{2} \times 2 - \frac{3}{24} \times 2^3 \right]$$

$$= 6$$

Center of mass $(\bar{x}, \bar{y})$

$$\bar{x} = \frac{1}{m} \iint_D x \rho(x, y) dA$$

$$= \frac{1}{6} \int_0^2 \int_{\frac{x}{2}}^{3-x} [x^2 + xy] dy dx$$

$$\begin{aligned}
&= \frac{1}{6} \int_0^2 \left[xy + \frac{1}{2} xy^2 \right]_{\frac{y}{2}}^{3-x} dx \\
&= \frac{1}{6} \int_0^2 \left\{ x^2(3-x) + \frac{1}{2} x(3-x)^2 - x^2\left(\frac{x}{2}\right) - \frac{1}{2} x\left(\frac{x}{2}\right)^2 \right\} dx \\
&= \frac{1}{6} \int_0^2 \left\{ 3x^2 - x^3 - \frac{x^3}{2} + \frac{1}{2} x(9 - 6x + x^2) - \frac{1}{2} x \cdot \frac{1}{4} x^2 \right\} dx \\
&= \frac{1}{6} \int_0^2 \left(3x^2 - x^3 - \frac{x^3}{2} + \frac{9}{2}x - 3x^2 + \frac{1}{2}x^3 - \frac{1}{8}x^3 \right) dx \\
&= \frac{1}{6} \int_0^2 \left(\frac{9}{2}x - \frac{1}{8}x^3 \right) dx \\
&= \frac{1}{6} \left[\frac{9}{2} \times \frac{1}{2} x^2 - \frac{1}{8} \times \frac{1}{4} x^4 \right]_0^2 \\
&= \frac{1}{6} \left[\frac{9 \times 4}{4} - \frac{1 \times 16}{24} \right] \\
&= \frac{2}{3}
\end{aligned}$$

$$\begin{aligned}
\bar{y} &= \frac{1}{m} \iint_D y \cdot \rho(x, y) dA \\
&= \frac{1}{6} \int_0^2 \int_{\frac{y}{2}}^{3-x} (xy + y^2) dy dx \\
&= \frac{1}{6} \int_0^2 \left[\frac{1}{2} xy^2 + \frac{1}{3} y^3 \right]_{\frac{y}{2}}^{3-x} dx \\
&= \frac{1}{6} \int_0^2 \left[\frac{1}{2} x \left\{ (3-x)^2 - \left(\frac{x}{2}\right)^2 \right\} + \frac{1}{3} \left\{ (3-x)^3 - \left(\frac{x}{2}\right)^3 \right\} \right] dx \\
&= \frac{1}{6} \int_0^2 \left[\frac{1}{2} x \left(9 - 6x + x^2 - \frac{x^2}{4} \right) + \frac{1}{3} \left(27 - 27x + 9x^2 - x^3 - \frac{x^3}{8} \right) \right] dx \\
&= \frac{1}{6} \int_0^2 \left[\frac{1}{2} 9x - 3x^2 + \frac{3}{8} x^3 + 9 - 9x + 3x^2 - \frac{3}{8} x^3 \right] dx \\
&= \frac{1}{6} \int_0^2 (9 - \frac{3}{2}x) dx
\end{aligned}$$

$$= \frac{1}{6} \left[0x - \frac{2}{2} \times \frac{1}{2} x^2 \right]_0^2$$

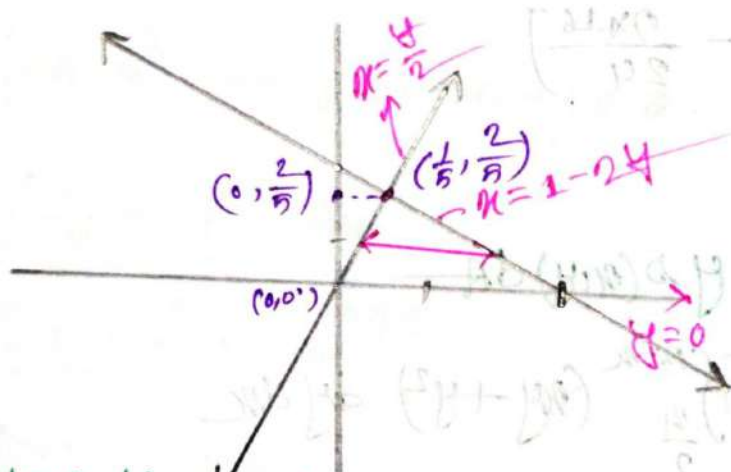
$$= \frac{1}{6} \left(18 - \frac{2}{4} \times 4 \right)$$

$$= \frac{3}{2}$$

$\therefore$ mass $m = 6$

Center of mass $(\bar{x}, \bar{y}) = \left(\frac{3}{4}, \frac{3}{2} \right)$

⑥ D is the triangular region enclosed by the lines $y=0$, $y=2x$, $x+2y=1$; $\rho(x,y)=x$



Find the intersection point:

$$\frac{y}{2} = 1 - 2y$$

$$y = 2 - 4y$$

$$5y = 2$$

$$y = \frac{2}{5}, x = \frac{1}{5}; (\bar{x}, \bar{y}) = \left(\frac{1}{5}, \frac{2}{5} \right)$$

We know:-

$$m = \iint_D \rho(x, y) dA$$

$$= \int_0^{\frac{2}{5}} \int_{\frac{y}{2}}^{1-2y} \rho \, dx \, dy$$

$$= \int_0^{\frac{2}{5}} \left[\frac{1}{2} \rho x^2 \right]_{\frac{y}{2}}^{1-2y} dy$$

$$= \int_0^{\frac{2}{5}} \frac{1}{2} \left\{ (1-2y)^2 - \left(\frac{y}{2}\right)^2 \right\} dy$$

$$= \frac{1}{2} \int_0^{\frac{2}{5}} \left(1 - 4y + 4y^2 - \frac{y^2}{4} \right) dy$$

$$= \frac{1}{2} \left[y - 4 \times \frac{1}{2} y^2 + \frac{4y^3}{3} - \frac{1}{4} \times \frac{1}{3} y^3 \right]_0^{\frac{2}{5}}$$

$$= \frac{1}{2} \left\{ \frac{2}{5} - 2 \left(\frac{2}{5} \right)^2 + \frac{16}{12} \left(\frac{2}{5} \right)^3 \right\}$$

$$= \frac{2}{25}$$

$$\bar{x} = \frac{1}{m} \iint_D x \rho(x, y) dA$$

$$= \frac{25}{2} \int_0^{\frac{2}{5}} \int_{\frac{y}{2}}^{1-2y} x^2 \rho \, dx \, dy$$

$$= \frac{25}{2} \int_0^{\frac{2}{5}} \left[\frac{1}{3} x^3 \right]_{\frac{y}{2}}^{1-2y} dy$$

$$= \frac{25}{2} \int_0^{\frac{2}{5}} \frac{1}{3} \left\{ (1-2y)^3 - \left(\frac{y}{2}\right)^3 \right\} dy$$

$$= \frac{25}{2 \times 3} \int_0^{\frac{2}{5}} \left(1 - 6y + 12y^2 - 8y^3 - \frac{y^3}{8} \right) dy$$

$$= \frac{25}{6} \left[y - 6 \times \frac{1}{2} y^2 + \frac{12y^3}{3} - \frac{8y^4}{4} - \frac{1}{8} \times \frac{1}{4} y^4 \right]_0^{\frac{2}{5}}$$

$$= \frac{25}{6} \left[\frac{2}{5} - 3 \left(\frac{2}{5} \right)^2 + \frac{12}{3} \left(\frac{2}{5} \right)^3 - \frac{65}{32} \left(\frac{2}{5} \right)^4 \right] = \frac{31}{60}$$

$$\bar{y} = \frac{1}{m} \iint_D y \rho(x, y) dA$$

$$= \frac{25}{2} \int_0^{\frac{2}{5}} \int_{\frac{y}{2}}^{1-2y} xy \cdot dm dy$$

$$= \frac{25}{2} \int_0^{\frac{2}{5}} \left(\frac{1}{2} x y^2 \right)_{\frac{y}{2}}^{1-2y} dy$$

$$= \frac{25}{4} \int_0^{\frac{2}{5}} y \left[(1-2y)^2 - \left(\frac{y}{2} \right)^2 \right] dy$$

$$= \frac{25}{4} \int_0^{\frac{2}{5}} \left(y - 4y^2 + 4y^3 - \frac{y^3}{4} \right) dy$$

$$= \frac{25}{4} \left[\frac{1}{2} y^2 - 4 \times \frac{1}{3} y^3 + \frac{15}{4} y^4 - \frac{1}{4} y^4 \right]_0^{\frac{2}{5}}$$

$$= \frac{25}{4} \left[\frac{1}{2} \left(\frac{2}{5} \right)^2 - \frac{4}{3} \left(\frac{2}{5} \right)^3 + \frac{15}{4} \left(\frac{2}{5} \right)^4 - \frac{1}{4} \left(\frac{2}{5} \right)^4 \right]$$

$$= \frac{25}{4} \left(1 - \frac{7}{375} \right)$$

$$= \frac{7}{60}$$

$$\therefore \text{mass } m = \frac{2}{25}$$

$$\text{Center of mass } (\bar{x}, \bar{y}) = \left(\frac{21}{60}, \frac{7}{60} \right)$$

⑦ D is bounded by $y = 1 - x^2$ and $y = 0$, $\rho(x, y) = ky$

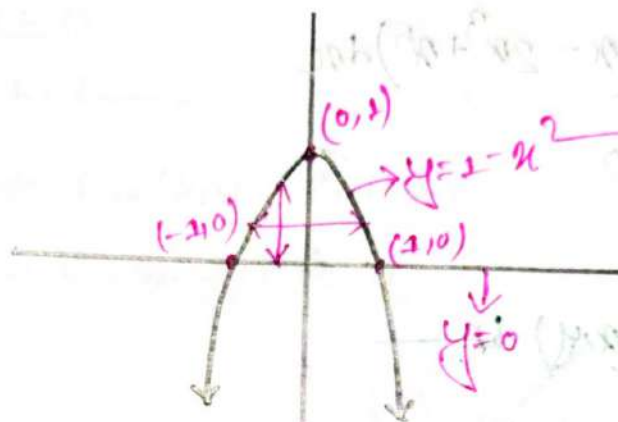
⇒ Find the intersection point:—

$$1 - x^2 = 0$$

$$x = \pm 1$$

$$x = 1, y = 0; (x, y) = (1, 0)$$

$$x = -1, y = 0; (x, y) = (-1, 0)$$



we know:—

$$m = \iint_D \rho(x, y) \, dA$$

$$= \int_{-1}^1 \int_0^{1-x^2} ky \, dy \, dx$$

$$= \int_{-1}^1 \left[k \frac{1}{2} y^2 \right]_0^{1-x^2} dx$$

$$= \frac{k}{2} \int_{-1}^1 (1-x^2)^2 \cdot dx$$

$$= \frac{k}{2} \int_{-1}^1 (1 - 2x^2 + x^4) \cdot dx$$

$$= k \left[x - 2x \frac{1}{3} + \frac{1}{5} x^5 \right]_{-1}^1$$

$$= k \left[1 - \frac{2}{3} + \frac{1}{5} \right] = \frac{8}{15} k$$

$$\bar{x} = \frac{1}{m} \iint_D xky \, dA$$

$$= \frac{15}{8k} \int_{-1}^1 \int_0^{1-x^2} xy \cdot dy \, dx$$

$$= \frac{15}{8} \int_{-1}^1 \left[\frac{1}{2} x y^2 \right]_0^{1-x^2} dx$$

$$= \frac{15}{8 \times 2} \int_{-1}^1 x (1-x^2)^2 dx$$

$$= \frac{15}{16} \int_{-1}^1 (x - 2x^3 + x^5) dx$$

$$= \frac{15}{16} \times 0$$

$$= 0$$

$$\bar{y} = \frac{1}{m} \iint_D ykx \, dA$$

$$= \frac{15}{8k} \int_{-1}^1 \int_0^{1-x^2} ky^2 \cdot dy \, dx$$

$$= \frac{15}{8} \int_{-1}^1 \left[\frac{1}{3} y^3 \right]_0^{1-x^2} dx$$

$$= \frac{15}{24} \int_{-1}^1 (1-x^2)^3 dx$$

$$= \frac{15}{24} \int_{-1}^1 (1 - 3x^2 + 3x^4 - x^6) dx$$

$$= \frac{5}{4} \left[x - \frac{1}{3} x^3 + 3 \times \frac{1}{5} x^5 - \frac{1}{7} x^7 \right]_{-1}^1$$

$$= \frac{5}{4} \left(1 - 1 + \frac{3}{5} - \frac{1}{7} \right)$$

$$= \frac{4}{7}$$

$$\therefore \text{mass } m = \frac{8K}{18}$$

$$\text{Center of mass } (\bar{x}, \bar{y}) = \left(0, \frac{4}{2}\right) \cdot \left(\frac{10}{2} - \frac{10}{2} + \frac{10}{2}\right) = \frac{10}{2}$$

⑧ D is bounded by the $y = x + 2$ and $y = x^2$; $\rho(x, y) = 1$

Find the intersection point:

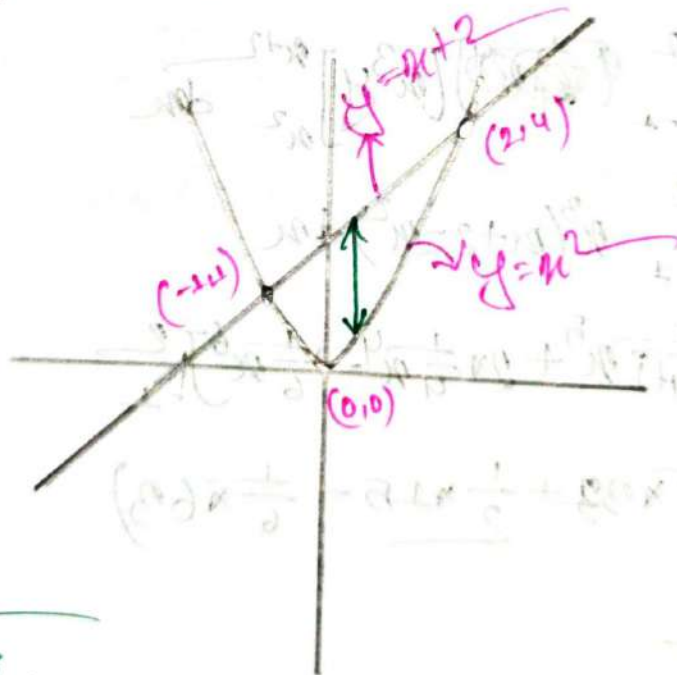
$$x + 2 = x^2$$

$$\Rightarrow x^2 - x - 2 = 0$$

$$\Rightarrow x = -1, 2$$

$$\therefore x = -1, y = 1 \rightarrow (x, y) = (-1, 1)$$

$$x = 2, y = 4 \rightarrow (x, y) = (2, 4)$$



We know:

$$m = \iint_D \rho(x, y) \, dA$$

$$= \int_{-1}^2 \int_{x^2}^{x+2} 1 \, dy \, dx$$

$$= \int_{-1}^2 \left[y \right]_{x^2}^{x+2} dx$$

$$= \int_{-1}^2 K x^2 (x+2-x^2) dx$$

$$= \int_{-1}^2 K (x^3 + 2x^2 - x^4) \cdot dx$$

$$= K \left[\frac{1}{4} x^4 + 2x \frac{1}{3} x^3 - \frac{1}{5} x^5 \right]_{-1}^2$$

$$= K \left[\frac{1}{4} \cdot 16 + \frac{2}{3} \cdot 8 - \frac{1}{5} \cdot 32 \right]$$

$$= \frac{63}{20} K$$

$$\bar{x} = \frac{1}{m} \iint_D x \rho(x, y) dA$$

$$= \frac{1}{\frac{63K}{20}} \int_{-1}^2 \int_{x^2}^{x+2} K x^3 \cdot dy dx$$

$$= \frac{20K}{63K} \int_{-1}^2 \left[x^3 y \right]_{x^2}^{x+2} dx$$

$$= \frac{20}{63} \int_{-1}^2 x^3 (x+2-x^2) dx$$

$$= \frac{20}{63} \left[\frac{1}{4} x^5 + 2x \frac{1}{4} x^4 - \frac{1}{6} x^6 \right]_{-1}^2$$

$$= \frac{20}{63} \left(\frac{1}{8} \cdot 32 + \frac{1}{2} \cdot 16 - \frac{1}{6} \cdot 63 \right)$$

$$= \frac{8}{7}$$

$$\bar{y} = \frac{1}{m} \iint_D y \rho(x, y) \cdot dA$$

$$= \frac{20}{63K} \int_{-1}^2 \int_{x^2}^{x+2} K y x^2 \cdot dy dx$$

$$= \frac{20K}{63K} \int_{-1}^2 x^2 \frac{1}{2} [y^2]_{x^2}^{x+2} \cdot dx$$

$$= \frac{20}{63} \int_{-1}^2 \left[x^2 \times \frac{1}{2} \right] \{ (x+2)^2 - x^4 \} dx$$

$$= \frac{20}{63 \times 2} \int_{-1}^2 x^2 (x^2 + 2x \cdot 2 + 4 - x^4) \cdot dx$$

$$= \frac{10}{63} \int_{-1}^2 (x^4 + 4x^3 + 4x^2 - x^6) dx$$

$$= \frac{10}{63} \left[\frac{1}{5} x^5 + \frac{1}{4} \times 4x^4 + 4 \times \frac{1}{3} x^3 - \frac{1}{7} x^7 \right]_{-1}^2$$

$$= \frac{10}{63} \left(\frac{1}{5} \times 33 + 16 + \frac{4}{3} \times 0 - \frac{1}{7} \times 128 \right)$$

$$= \frac{118}{40}$$

$$\therefore \text{mass } m = \frac{63}{20} \text{ kg}$$

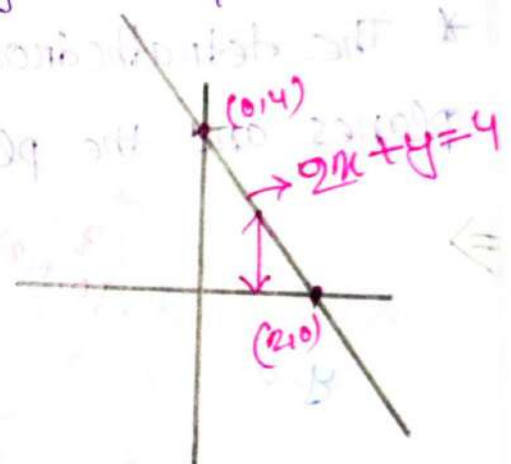
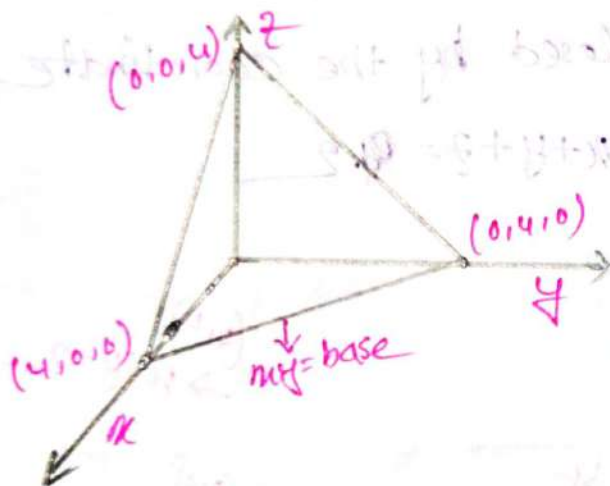
$$\text{Center of mass } (x, y) = \left(\frac{8}{7}, \frac{118}{40} \right)$$

Triple Integral Application

Pages 1038!

10-228 Use a triple integral to find the volume of the given solid.

- (10) The tetrahedron enclosed by the coordinate planes and the plane $2x + y + z = 4$.



⇒ we know

$$\begin{aligned}
 V &= \iiint_D dv \\
 &= \int_0^2 \int_0^{4-2x} \int_0^{4-2x-y} dz dy dx \\
 &= \int_0^2 \int_0^{4-2x} \left[z \right]_0^{4-2x-y} dy dx \\
 &= \int_0^2 \int_0^{4-2x} (4-2x-y) dy dx \\
 &= \int_0^2 \left[4y - 2xy - \frac{1}{2}y^2 \right]_0^{4-2x} dx
 \end{aligned}$$

$$= \int_0^2 \left[4(4-2x) - 2x(4-2x) - \frac{1}{2}(4-2x)^2 \right] \cdot dx$$

$$= \int_0^2 (16 - 8x - 8x + 4x^2 - 2x4 + 8x - 2x^2) dx$$

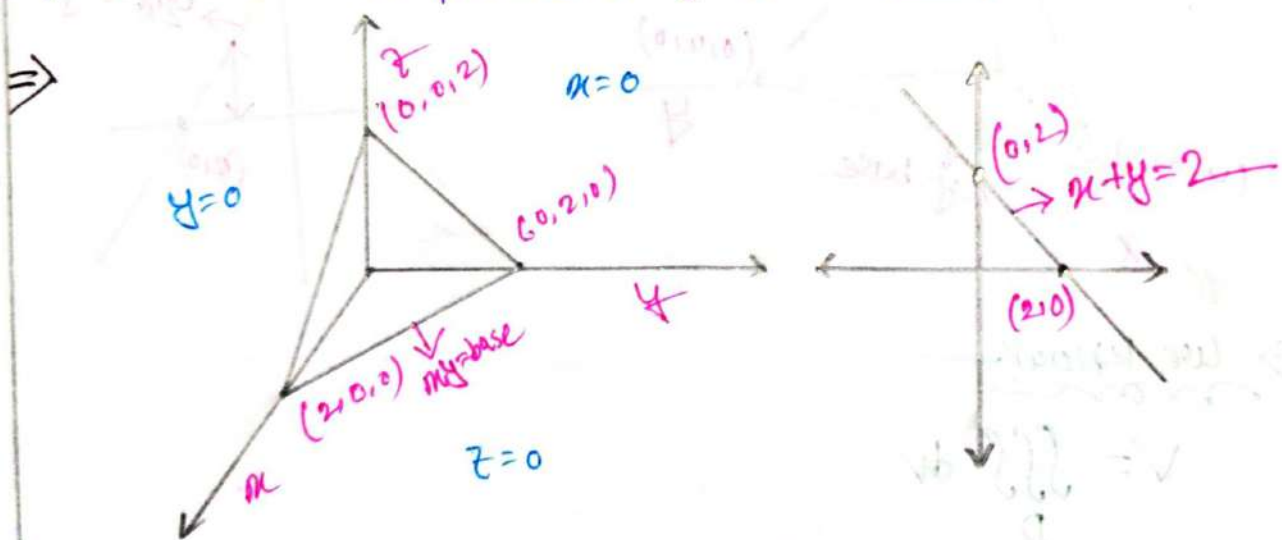
$$= \int_0^2 (8 - 8x + 2x^2) dx$$

$$= \left[8x - \frac{1}{2} 8x^2 + \frac{2}{3} x^3 \right]_0^2$$

$$= 16 - 4 \times 4 + \frac{2}{3} \times 8$$

$$= \frac{16}{3}$$

* The tetrahedron enclosed by the coordinate planes and the plane $x+y+z=2$



We know —
volume

$$V = \iiint_D dV$$

$$= \int_0^2 \int_0^{2-x} \int_0^{2-x-y} dz dy dx$$

$$= \int_0^2 \int_0^{2-x} [z]_0^{2-x-y} dy dx$$

$$= \int_0^2 \int_0^{2-x} (2-x-y) dy \cdot dx$$

$$= \int_0^2 [2y - xy - \frac{1}{2}y^2]^{2-x} dx$$

$$= \int_0^2 [2(2-x) - x(2-x) - \frac{1}{2}(2-x)^2] dx$$

$$= \int_0^2 [4 - 2x - 2x + x^2 - \frac{1}{2}(4 - 2(x+x^2))] dx$$

$$= \int_0^2 [4 - 2x - 2x + x^2 - 2 + 2x - \frac{1}{2}x^2] dx$$

$$= \int_0^2 [2 - 2x + \frac{1}{2}x^2] dx$$

$$= [2x - \frac{1}{2} \times 2x^2 + \frac{1}{2} \times \frac{1}{3}x^3]_0^2$$

$$= 4 - 4 + \frac{8}{6}$$

$$= \frac{8}{6}$$

(20) The solid enclosed by the paraboloids $y = x^2 + z^2$ and $y = 8 - x^2 - z^2$

⇒ now, we re-write the equation:

$$y = (x^2 + z^2), \quad y = 8 - (x^2 + z^2)$$

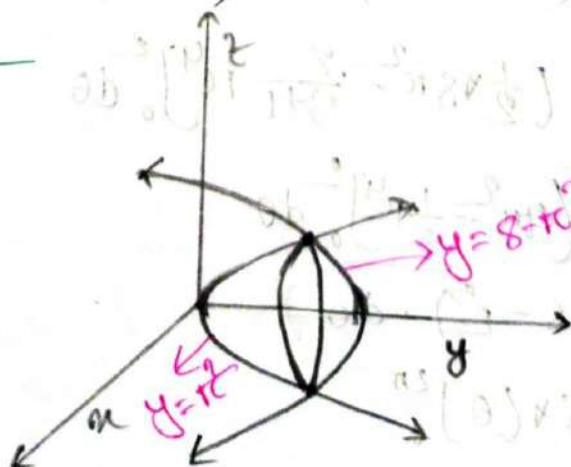
⇒ Solve the equation:

$$x^2 + z^2 = 8 - (x^2 + z^2)$$

$$2x^2 + 2z^2 = 8$$

$$x^2 + z^2 = 4$$

$$x^2 + z^2 = 2^2$$



we know:

$$x = r \cos \theta$$

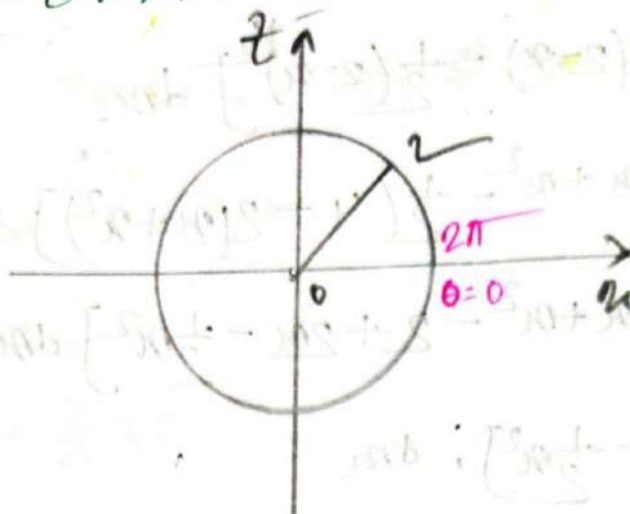
$$y = r \sin \theta$$

$$x^2 + z^2 = r^2$$

$$y = r^2$$

$$y = 8 - r^2$$

Find the polar coordinates:



We know:

$$V = \iiint_D dV$$

$$= \int_0^{2\pi} \int_0^2 \int_{r^2}^{8-r^2} dy \, r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 \left[y \right]_{r^2}^{8-r^2} r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 (8 - r^2 - r^2) r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 (8r - 2r^3) \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[\frac{1}{2} 8r^2 - \frac{2}{3+1} r^4 \right]_0^2 d\theta$$

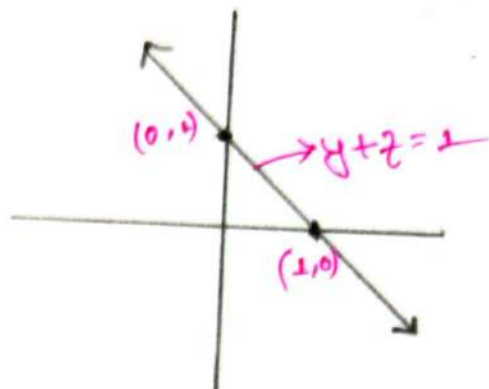
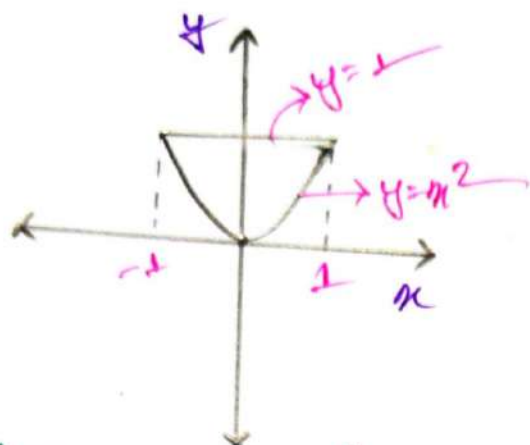
$$= \int_0^{2\pi} \left[4r^2 - \frac{1}{2} r^4 \right]_0^2 d\theta$$

$$= \int_0^{2\pi} (16 - 2) \, d\theta$$

$$= 8 \times [\theta]_0^{2\pi}$$

$$= 16\pi$$

② The solid enclosed by the cylinder $y=x^2$ and the planes $z=0$, and $y+z=1$

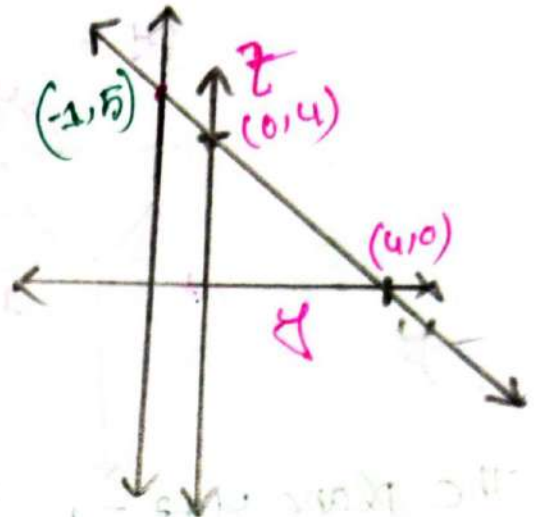
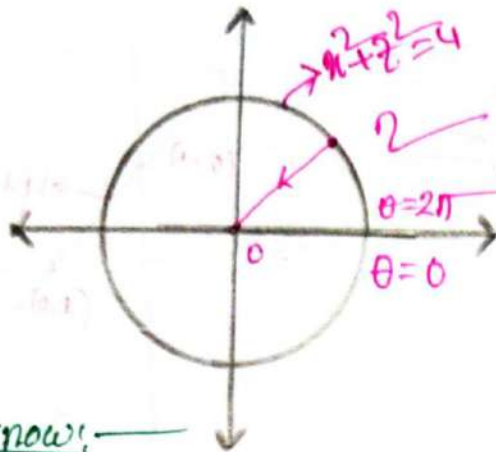


The plane $y+z=1$ intersects the xy plane in the line $y=1$

We know:

$$\begin{aligned}
 V &= \iiint_D dv \\
 &= \int_{-1}^1 \int_{x^2}^1 \int_0^{1-y} dz dy dx \\
 &= \int_{-1}^1 \int_{x^2}^1 [z]_0^{1-y} dy dx \\
 &= \int_{-1}^1 \left[y - \frac{1}{2} y^2 \right]_{x^2}^1 dx \\
 &= \int_{-1}^1 \left\{ 1 - x^2 - \left(\frac{1}{2} x^2 - \frac{1}{2} x^4 \right) \right\} dx \\
 &= \int_{-1}^1 \left(1 - x^2 - \frac{1}{2} x^2 + \frac{1}{2} x^4 \right) dx \\
 &= 2 \left[\frac{1}{2} x - \frac{1}{3} x^3 + \frac{1}{2} x \frac{1}{5} x^5 \right]_0^1 \\
 &= 2 \times \left(\frac{1}{2} - \frac{1}{3} + \frac{1}{10} \right) \\
 &= \frac{8}{15}
 \end{aligned}$$

② The solid enclosed by the cylinder $x^2 + z^2 = 4$ and the planes $y = -1$ and $y + z = 4$



We know:—

$$x = r \cos \theta$$

$$z = r \sin \theta$$

in given:—

$$y + z = 4$$

$$\therefore y = 4 - z$$

$$= 4 - r \sin \theta$$

$$\therefore V = \iiint dv$$

$$= \int_0^{2\pi} \int_0^2 \int_{-1}^{4-r\sin\theta} dy \, r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 [y]_{-1}^{4-r\sin\theta} \cdot \pi \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 [4 - r\sin\theta + 1] r \, dr \, d\theta$$

$$= \int_0^{2\pi} \left[5 \times \frac{1}{2} r^2 + \frac{1}{3} r^3 \sin\theta \right]_0^2 d\theta$$

$$= \int_0^{2\pi} \left[\frac{5}{2} \times 4 + \frac{8}{3} \sin\theta \right] d\theta$$

$$= [10\theta]_0^{2\pi} + \frac{8}{3} (\cos 2\pi - \cos 0)$$

$$= [10\theta]_0^{2\pi} + \frac{8}{3} - \frac{8}{3}$$

$$= 20\pi$$

Ans:

Differential Equations

Chap 1

Order & Degree:-

- ① $\frac{dy}{dx} = f(x, y) \Rightarrow 1^{\text{st}} \text{ order } 1^{\text{st}} \text{ degree.}$
- ② $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 2 \Rightarrow 2^{\text{nd}} \text{ order } 1^{\text{st}} \text{ degree.}$
- ③ $\left(\frac{d^2y}{dx^2}\right)^2 - 2\frac{dy}{dx} = \sin x \Rightarrow 2^{\text{nd}} \text{ order } 2^{\text{nd}} \text{ degree.}$

First Order Differential equations:-

$\Rightarrow$ An Ordinary differential equation can be written as $\frac{dy}{dx} = f(x, y)$

Methods of solution:-

- ① Separation of variable / Variable separable.

$$\Rightarrow \frac{dy}{dx} = f(x) \cdot g(y)$$

$$\Rightarrow \frac{dy}{g(y)} = f(x) \cdot dx$$

Ex 8.1

$$\textcircled{1} \frac{dy}{dx} = x \cdot y$$

$$\Rightarrow \frac{dy}{y} = x \cdot dx$$

Q. Integrating factor

$$\Rightarrow \frac{dy}{dx} + P(x)y = Q(x) \quad \text{--- (1)}$$

i) Find the integrating factor. IF/IF(x)

$$\text{IF} = \text{IF}(x) = e^{\int P(x) \cdot dx}$$

ii) Multiply equation (1) by IF(x)

iii) Equation (1) can be written as

$$\frac{d}{dx} \{ y \cdot \text{IF}(x) \} = Q(x) \cdot \text{IF}(x)$$

iv) Using integration on we get the general solution as

$$y \cdot \text{IF}(x) = \int Q(x) \cdot \text{IF}(x) \cdot dx$$

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Find the general solution of the following differential equation.

5. $\frac{dy}{dt} = (1+y)^2$

$$\Rightarrow \frac{dy}{dt} = 1 + 2y + y^2$$

$$\Rightarrow \frac{dy}{y^2} = \frac{1}{y^2} dt$$

$$\Rightarrow \int \frac{dy}{y^2} = \int \frac{1}{y^2} dt$$

$$\Rightarrow -\frac{1}{y} = \frac{1}{3} t^3 + C$$

$$\Rightarrow \frac{1}{y} = -\frac{1}{3} t^3 - C$$

$$\Rightarrow y = -\frac{3}{t^3} + A$$

6. $\frac{dy}{dt} = t^4 y$

$$\Rightarrow \frac{dy}{y} = t^4 dt$$

$$\Rightarrow \int \frac{dy}{y} = \int t^4 dt$$

$$\Rightarrow \ln(y) = \frac{1}{5} t^5 + C$$

$$\Rightarrow e^{\ln(y)} = e^{\frac{1}{5} t^5 + C}$$

$$\Rightarrow y = e^{\frac{1}{5} t^5} \cdot e^C$$

$$\Rightarrow y = e^{\frac{1}{5} t^5} \cdot A$$

7. $\frac{dy}{dt} = 2y + 1$

$$\Rightarrow \frac{dy}{2y+1} = dt$$

$$\Rightarrow \int \frac{dy}{2y+1} = \int dt$$

$$\Rightarrow \frac{1}{2} \ln(2y+1) = t + C$$

$$\Rightarrow \ln(2y+1) = 2t + 2C$$

$$\Rightarrow e^{\ln(2y+1)} = e^{2t+2C}$$

$$\Rightarrow 2y = e^{2t} \cdot e^C - 1$$

$$\Rightarrow y = \frac{1}{2} e^{2t} \cdot A - \frac{1}{2}$$

$$\textcircled{8} \quad \frac{dy}{dt} = 2-y$$

$$\Rightarrow \frac{dy}{2-y} = dt$$

$$\Rightarrow \int \frac{dy}{2-y} = \int dt$$

$$\Rightarrow \ln|2-y| = t + c$$

$$\Rightarrow e^{\ln(2-y)} = e^{t+c}$$

$$\Rightarrow 2-y = e^t \cdot e^c$$

$$\Rightarrow y = 2 - Ae^t$$

$$\textcircled{9} \quad \frac{dy}{dt} = e^{-y}$$

$$\Rightarrow \frac{dy}{e^{-y}} = dt$$

$$\Rightarrow \int \frac{dy}{e^{-y}} = \int dt$$

$$\Rightarrow \int e^y = \int dt$$

$$\Rightarrow e^y = t + c$$

$$\Rightarrow \ln(e^y) = \ln(t+c)$$

$$\Rightarrow y = \ln(t+c)$$

$$\textcircled{10} \quad \frac{dx}{dt} = 1+x^2$$

$$\Rightarrow \frac{dx}{1+x^2} = dt$$

$$\Rightarrow \int \frac{1}{1+x^2} dx = \int dt$$

$$\Rightarrow \tan^{-1} x = t + c$$

$$\Rightarrow x = \tan(t+c)$$

$$\textcircled{11} \quad \frac{dy}{dt} = 2+y^2+3y^2$$

$$\Rightarrow \frac{dy}{dt} = y^2(2t+3)$$

$$\Rightarrow \frac{dy}{y^2} = (2t+3) dt$$

$$\Rightarrow \int \frac{dy}{y^2} = \int (2t+3) dt$$

$$\Rightarrow -\frac{1}{y} = 2 \times \frac{1}{2} t^2 + 3t + c$$

$$\Rightarrow y = -\frac{1}{t^2 - \frac{1}{3}t + c}$$

$$(12) \frac{dy}{dt} = \frac{t}{y}$$

$$\Rightarrow y dy = t dt$$

$$\Rightarrow \int y dy = \int t dt$$

$$\Rightarrow \frac{1}{2} y^2 = \frac{1}{2} t^2$$

$$\Rightarrow y = \pm \sqrt{t^2 + c}$$

$$(13) \frac{dy}{dt} = \frac{t}{t^2 y + y}$$

$$\Rightarrow \frac{dy}{dt} = \frac{t}{y(t^2 + 1)}$$

$$\Rightarrow y \cdot dy = \frac{t}{t^2 + 1} \cdot dt$$

$$\Rightarrow \int y \cdot dy = \int \frac{t}{t^2 + 1} \cdot dt$$

$$\Rightarrow \frac{1}{2} y^2 = \frac{1}{2} \ln|t^2 + 1| + c$$

$$\Rightarrow y = \pm \sqrt{\ln(t^2 + 1) + c}$$

$$(14) \frac{dy}{dt} = t \sqrt[3]{y}$$

$$\Rightarrow \int \frac{dy}{y^{\frac{1}{3}}} = \int t \cdot dt$$

$$\Rightarrow \frac{3}{2} y^{\frac{2}{3}} = \frac{1}{2} t^2 + c$$

$$\Rightarrow y^{\frac{2}{3}} = \frac{2}{3} \times \frac{1}{2} t^2 + \frac{2}{3} c$$

$$\Rightarrow y^{\frac{2}{3}} = \frac{1}{3} t^2 + c_1$$

$$(15) \frac{dy}{dt} = \frac{1}{2y+1}$$

$$\Rightarrow (2y+1) dy = dt$$

$$\Rightarrow \int (2y+1) dy = \int dt$$

$$\Rightarrow 2 \times \frac{1}{2} y^2 + y = t + c$$

$$\Rightarrow y^2 + y = t + c$$

$$=$$

$$(16) \frac{dy}{dt} = \frac{2y+1}{t}$$

$$\Rightarrow \frac{dy}{2y+1} = \frac{dt}{t}$$

$$\Rightarrow \int \frac{dy}{2y+1} = \int \frac{dt}{t}$$

$$\Rightarrow \frac{1}{2} \ln|2y+1| = \ln|t| + c$$

$$\Rightarrow \ln(2y+1)^{\frac{1}{2}} - \ln(t) = c$$

$$\Rightarrow \ln\left(\frac{(2y+1)^{\frac{1}{2}}}{t}\right) = c$$

$$\Rightarrow e^{\ln\left(\frac{(2y+1)^{\frac{1}{2}}}{t}\right)} = e^c$$

$$\Rightarrow (2y+1)^{\frac{1}{2}} = At$$

$$\Rightarrow 2y+1 = (At)^2$$

$$\Rightarrow y = \frac{(A^2 t^2 - 1)}{2}$$

$$(17) \frac{dy}{dt} = y(1-y)$$

$$\Rightarrow \frac{dy}{y(1-y)} = dt$$

$$\text{Let, } \frac{1}{y(1-y)} = \frac{A}{y} + \frac{B}{1-y}$$

$$\therefore A = \frac{1}{1-y} \text{ at } y=0 \Rightarrow A=1$$

$$\therefore y=1; B = \frac{1}{y} = \frac{1}{1-y} = 1$$

$$\therefore \int \left(\frac{1}{y} + \frac{1}{1-y} \right) dy = \int dt$$

$$\Rightarrow \ln(y) + (-\ln(1-y)) = t + C$$

$$\Rightarrow \ln y - \ln(1-y) = t + C$$

$$\Rightarrow \ln y - \ln(1-y) = t + C$$

$$\Rightarrow e^{\ln\left(\frac{y}{1-y}\right)} = e^{t+C}$$

$$\Rightarrow \frac{y}{1-y} = Ae^t$$

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$$(18) \frac{dy}{dt} = \frac{4t}{1+3y^2}$$

$$\Rightarrow (1+3y^2)dy = 4t dt$$

$$\Rightarrow \int (1+3y^2)dy = \int 4t \cdot dt$$

$$\Rightarrow y + 3 \times \frac{1}{3} y^3 = 4 \times \frac{1}{2} t^2 + C$$

$$\Rightarrow y + y^3 = 2t^2 + C$$

$\Rightarrow$

$$(19) \frac{dv}{dt} = t^2 v - 2 - 2v + t^2$$

$$\Rightarrow \frac{dv}{dt} = v(t^2 - 2) + 1(t^2 - 2)$$

$$\Rightarrow \frac{dv}{dt} = (v+1)(t^2-2)$$

$$\Rightarrow \frac{dv}{v+1} = (t^2-2) dt$$

$$\Rightarrow \int \frac{dv}{v+1} = \int (t^2-2) dt$$

$$\Rightarrow \ln(v+1) = \frac{1}{3}t^3 - 2t + c$$

$$\Rightarrow e^{\ln(v+1)} = e^{\frac{1}{3}t^3 - 2t + c}$$

$$\Rightarrow (v+1) = e^{\frac{1}{3}t^3 - 2t} \cdot e^c$$

$$\Rightarrow v = e^{\frac{1}{3}t^3 - 2t} \cdot A - 1$$

$$\therefore v = Ae^{\frac{1}{3}t^3 - 2t} - 1$$

$$\textcircled{20.} \quad \frac{dy}{dt} = \frac{1}{t+y+t+y+1}$$

$$\Rightarrow \frac{dy}{dt} = \frac{1}{y(t+y)+1(t+y)}$$

$$\Rightarrow \frac{dy}{dt} = \frac{1}{(y+1)(t+y)}$$

$$\Rightarrow \frac{(y+1)dy}{1} = \frac{dt}{(t+y)}$$

$$\Rightarrow \int (y+1) dy = \int \frac{1}{t+y} \cdot dt$$

$$\Rightarrow \frac{1}{2}y^2 + y = \ln(t+y) + c$$

$$(21) \quad \frac{dy}{dt} = \frac{e^t y}{1+y^2}$$

$$\Rightarrow \frac{(1+y^2) dy}{y} = e^t \cdot dt$$

$$\Rightarrow \int \frac{1+y^2}{y} dy = \int e^t \cdot dt$$

$$\Rightarrow \ln(y) + \frac{1}{2}y^2 = e^t + C$$

$$(22) \quad \frac{dy}{dt} = y^2 - 4$$

$$\Rightarrow \frac{dy}{y^2 - 4} = dt$$

$$\Rightarrow \int \frac{dy}{y^2 - 2^2} = \int dt$$

$$\Rightarrow \frac{1}{2 \times 2} \ln \left| \frac{y-2}{y+2} \right| = t + C$$

$$\Rightarrow e^{\ln \left| \frac{y-2}{y+2} \right|} = e^{4t + 4C}$$

$$\Rightarrow \frac{y-2}{y+2} = e^{4t} \times A$$

$$\therefore \frac{y-2}{y+2} = A e^{4t}$$

Since:-

$$\int \frac{1}{x^2 - a^2} \cdot dx = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

$$(23) \quad \frac{dw}{dt} = \frac{w}{t}$$

$$\Rightarrow \frac{dw}{w} = \frac{dt}{t}$$

$$\Rightarrow \int \frac{dw}{w} = \int \frac{dt}{t}$$

$$\Rightarrow \ln(w) = \ln(t) + c$$

$$\Rightarrow e^{\ln(w)} = e^{\ln(t) + c}$$

$$\Rightarrow w = At$$

$$(24) \quad \frac{dy}{dx} = \sec y$$

$$\Rightarrow \frac{dy}{\sec y} = dx$$

$$\Rightarrow \int \frac{dy}{\sec y} = \int dx$$

$$\Rightarrow \int \frac{dy}{\frac{1}{\cos y}} = \int dx$$

$$\Rightarrow \int \cos y \cdot dy = \int dx$$

$$\Rightarrow \sin y = x + c$$

$$\Rightarrow y = \sin^{-1}(x + c)$$

Find the Solution of initial condition:-

(25) $\frac{dx}{dt} = -xt$; $x(0) = \frac{1}{\sqrt{\pi}}$ $\xrightarrow{\text{independent variable}}$ $\xrightarrow{\text{Dependent variable.}}$

$$\Rightarrow \frac{dx}{x} = -t \cdot dt$$

$$\Rightarrow \int \frac{dx}{x} = -\int t \, dt$$

$$\Rightarrow \ln(x) = -\frac{1}{2}t^2 + C \quad \text{--- (1)}$$

now:- Applying initial condition $x(0) = \frac{1}{\sqrt{\pi}}$ in eq (1) $\Rightarrow$
where $t=0$, $x = \frac{1}{\sqrt{\pi}}$

$$\therefore \ln\left(\frac{1}{\sqrt{\pi}}\right) = -\frac{1}{2}(0)^2 + C$$

$$\Rightarrow (\ln 1 - \ln \sqrt{\pi}) = C$$

$$\therefore C = -\ln \sqrt{\pi}$$

now. Substituting the value of C in the eq (1) $\Rightarrow$

$$\ln(x) = -\frac{1}{2}t^2 - \ln \sqrt{\pi}$$

$$\Rightarrow \ln(x) + \ln \sqrt{\pi} = -\frac{1}{2}t^2$$

$$\Rightarrow \ln(\sqrt{\pi} \cdot x) = -\frac{1}{2}t^2$$

$$\Rightarrow e^{\ln(\sqrt{\pi} \cdot x)} = e^{-\frac{1}{2}t^2}$$

$$\sqrt{\pi} \cdot x = e^{-\frac{1}{2}t^2}$$

$$\therefore x = \frac{1}{\sqrt{\pi}} e^{-\frac{1}{2}t^2}$$

$$(26) \frac{dy}{dt} = ty ; y(0) = 3 \quad \text{--- (1)}$$

$$\Rightarrow \frac{dy}{y} = t dt$$

$$\Rightarrow \int \frac{dy}{y} = \int t \cdot dt$$

$$\Rightarrow \ln(y) = \frac{1}{2} t^2 + c \quad \text{--- (1)}$$

now. Applying initial condition $y(0) = 3$ in eq (1) $\Rightarrow$
 where $t = 0, y = 3$

$$\therefore \ln(3) = c$$

putting the value of c in equation (1) $\Rightarrow$

$$\ln(y) = \frac{1}{2} t^2 + \ln(3)$$

$$\Rightarrow \ln(y) - \ln(3) = \frac{1}{2} t^2$$

$$\Rightarrow \ln\left(\frac{y}{3}\right) = \frac{1}{2} t^2$$

$$\Rightarrow e^{\ln\left(\frac{y}{3}\right)} = e^{\frac{1}{2} t^2}$$

$$\Rightarrow \frac{y}{3} = e^{\frac{1}{2} t^2}$$

$$\Rightarrow y = 3e^{\frac{1}{2} t^2}$$

$$(27) \frac{dy}{dt} = -y^2; y(0) = \frac{1}{2}$$

$$\Rightarrow \frac{dy}{y^2} = -dt$$

$$\Rightarrow \int \frac{dy}{y^2} = -\int dt$$

$$\Rightarrow -\frac{1}{y} = -t + c \quad \text{--- (1)}$$

Now Applying initial Condition $y(0) = \frac{1}{2}$ where
 $y = \frac{1}{2}, t = 0$

$$\therefore -\frac{1}{\frac{1}{2}} = c$$

$$\Rightarrow c = -2$$

putting the value of c in eq (1) $\Rightarrow$

$$-\frac{1}{y} = -t + (-2)$$

$$y = \frac{1}{t+2}$$

Ans:-

$$\textcircled{28} \quad \frac{dy}{dt} = +y^3, \quad y(0) = -1$$

$$\Rightarrow \frac{dy}{y^3} = +^2 dt$$

$$\Rightarrow \int \frac{dy}{y^3} = \int t^2 dt$$

$$\Rightarrow -\frac{1}{2} \times \frac{1}{y^2} = \frac{1}{3} t^3 + C \quad \text{--- (1)}$$

Now, Applying initial condition $y(0) = -1$ where
 $y = -1, t = 0$

$$-\frac{1}{2} \times \frac{1}{(-1)^2} = C$$

$$\therefore C = -\frac{1}{2}$$

Putting the value of C in equation (1) $\Rightarrow$

$$-\frac{1}{2} \times \left(\frac{1}{y^2} \right) = \frac{1}{3} t^3 + \left(-\frac{1}{2} \right)$$

$$\Rightarrow \frac{1}{y^2} = -\frac{2}{3} t^3 + \frac{2}{2}$$

$$\Rightarrow y = \sqrt{1 - \frac{2}{3} t^3}$$

$$\therefore y = \pm \sqrt{1 - \frac{2}{3} t^3}$$

$$(30) \frac{dy}{dt} = \frac{t}{y - t^2} ; y(0) = 4$$

$$\Rightarrow \frac{dy}{dt} = \frac{t}{y(1 - t^2)}$$

$$\Rightarrow y dy = \frac{t}{(1 - t^2)} \cdot dt$$

$$\Rightarrow \int y dy = \int \frac{t}{(1 - t^2)} \cdot dt$$

$$\Rightarrow \frac{1}{2} y^2 = -\frac{1}{2} \ln(1 - t^2) + c$$

$$\Rightarrow y^2 = -\ln(1 - t^2) + c \text{ ————— (1)}$$

now Applying initial condition $y(0) = 4$ | Where
 $y = 4, t = 0$

$$\therefore 16 = c$$

putting the value of c in eq (1) $\Rightarrow$

$$y = \pm \sqrt{-\frac{1}{2}(1 - t^2) + 16}$$

$$\therefore y = \pm \sqrt{16 - \frac{1}{2}(1 - t^2)}$$

Integrating Factors:

$$(1) \frac{dy}{dx} + P(x)y = Q(x)$$

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$$(1) \frac{dy}{dt} = \frac{-y}{t} + 2$$

$$\Rightarrow \frac{dy}{dt} + \frac{y}{t} = 2 \quad \text{--- (1)}$$

$\therefore$ Find the integrating factor, $I.F./ll(t) = e^{\int P(t) \cdot dt}$
 $= e^{\int \frac{1}{t} \cdot dt}$
 $= e^{\ln t}$
 $= t$

Multiply eq(1); by $ll(t) \Rightarrow$

$$t \cdot \frac{dy}{dt} + t \cdot \frac{y}{t} = 2t$$

$$\Rightarrow \frac{d}{dt} \{ t \cdot y \} = 2t$$

$$\Rightarrow d(ty) = 2t \cdot dt$$

$$\Rightarrow \int d(ty) = \int 2t \cdot dt$$

$$\Rightarrow ty = 2 \times \frac{1}{2} t^2 + C$$

$$\Rightarrow y = t + \frac{C}{t}$$

$$\textcircled{2} \quad \frac{dy}{dt} = \frac{3}{t}y + t^5$$

$$\Rightarrow \frac{dy}{dt} - \frac{3}{t}y = t^5 \quad \text{--- (1)}$$

$$\begin{aligned} \therefore I(t) &= e^{\int -\frac{3}{t} \cdot dt} \\ &= e^{-3 \ln(t)} \\ &= e^{\ln(t)^{-3}} \\ &= \frac{1}{t^3} \end{aligned}$$

multiply eq (1) by $I(t)$ $\Rightarrow$

$$\frac{1}{t^3} \cdot \frac{dy}{dt} - \frac{1}{t^3} \cdot \frac{3}{t}y = \frac{1}{t^3} t^5$$

$$\Rightarrow \frac{d}{dt} \left\{ \frac{1}{t^3} y \right\} = t^2$$

$$\Rightarrow d \left(\frac{1}{t^3} y \right) = t^2 \cdot dt$$

$$\Rightarrow \int d \left(\frac{1}{t^3} y \right) = \int t^2 \cdot dt$$

$$\Rightarrow \frac{1}{t^3} y = \frac{1}{3} t^3 + C$$

$$\Rightarrow y = \frac{1}{3} t^3 \cdot t^3 + C t^3$$

$$\therefore y = \frac{1}{3} t^6 + C t^3$$

$$(3) \quad \frac{dy}{dt} = - \frac{y}{1+t} + t^3$$

$$\Rightarrow \frac{dy}{dt} + \frac{y}{1+t} = t^3 \quad \text{--- (1)}$$

$$\therefore I.I(t) = e^{\int \frac{1}{1+t} \cdot dt} = e^{\ln(1+t)} = (1+t)$$

Multiply eq (1) by I.I(t) $\Rightarrow$

$$(1+t) \frac{dy}{dt} + (1+t) \cdot \frac{y}{1+t} = t^3(1+t)$$

$$\therefore \frac{d}{dt} \{ (1+t)y \} = t^3(1+t)$$

$$\Rightarrow d(1+t)y = t^3(1+t) dt$$

$$\Rightarrow \int d(1+t)y = \int t^3(1+t) dt$$

$$\Rightarrow (1+t)y = \frac{1}{4}t^4 + \frac{1}{5}t^5 + C$$

$$\Rightarrow y = \frac{1}{4} \times \frac{t^4}{(1+t)} + \frac{1}{5} \times \frac{t^5}{(1+t)} + \frac{C}{1+t}$$

$$y = \frac{1}{1+t} \left(\frac{1}{4}t^4 + \frac{1}{5}t^5 + C \right)$$

Ans: ---

$$(4) \frac{dy}{dt} = -2ty + 4e^{-t^2}$$

$$\Rightarrow \frac{dy}{dt} + 2ty = 4e^{-t^2} \quad \text{--- (1)}$$

$$\begin{aligned} \Rightarrow I(t) &= e^{\int 2t dt} \\ &= e^{2 \times \frac{1}{2} t^2} \\ &= e^{t^2} \end{aligned}$$

multiply eq (1) by $I(t)$

$$e^{t^2} \cdot \frac{dy}{dt} + e^{t^2} \cdot 2ty = 4e^{-t^2} \cdot e^{t^2}$$

$$\Rightarrow \frac{d}{dt} \{ e^{t^2} \cdot y \} = 4e^{-t^2} \cdot e^{t^2}$$

$$\Rightarrow d(e^{t^2} \cdot y) = 4e^{-t^2} \cdot e^{t^2} \cdot dt$$

$$\Rightarrow \int d(e^{t^2} \cdot y) = \int 4e^{-t^2} \cdot e^{t^2} \cdot dt$$

$$\Rightarrow e^{t^2} \cdot y = \int 4e^0 dt$$

$$\Rightarrow e^{t^2} \cdot y = 4t + C$$

$$\therefore y = \frac{1}{e^{t^2}} (4t + C)$$

Ans. —

$$(5) \frac{dy}{dt} - \frac{2t}{1+t^2} \cdot y = 3 \quad \text{--- (1)}$$

$$\begin{aligned} \Rightarrow I(t) &= e^{-\int \frac{2t}{1+t^2} \cdot dt} \\ &= e^{-\ln(1+t^2)} \\ &= e^{\ln(1+t^2)^{-1}} \\ &= \frac{1}{1+t^2} \end{aligned}$$

multiply eq (1) by $I(t) \Rightarrow$

$$\frac{1}{1+t^2} \cdot \frac{dy}{dt} - \frac{1}{1+t^2} \cdot \frac{2t}{1+t^2} \cdot y = 3 \times \frac{1}{1+t^2}$$

$$\Rightarrow \frac{d}{dt} \left(\frac{1}{1+t^2} \cdot y \right) = \frac{3}{1+t^2}$$

$$\Rightarrow \int d \left(\frac{1}{1+t^2} \right) y = \int \frac{3}{1+t^2} \cdot dt$$

$$\Rightarrow \left(\frac{1}{1+t^2} \right) y = 3 \tan^{-1}(t) + C$$

$$\Rightarrow y = (1+t^2) \{ 3 \tan^{-1}(t) + C \}$$

Ans:-

$$\textcircled{6} \quad \frac{dy}{dt} - \frac{2}{t}y = +3e^t \quad \text{---} \textcircled{1}$$

$$\begin{aligned} \therefore \mu(t) &= e^{-\int \frac{2}{t} \cdot dt} \\ &= e^{-2\ln(t)} \\ &= e^{\ln(t)^{-2}} \\ &= \frac{1}{t^2} \end{aligned}$$

Multiply eq (1) by $\mu(t) \Rightarrow$

$$\frac{1}{t^2} \cdot \frac{dy}{dt} - \frac{1}{t^2} \cdot \frac{2}{t}y = \frac{+3e^t}{+2}$$

$$\frac{d}{dt} \left(\frac{1}{t^2} y \right) = te^t$$

$$\Rightarrow \int d \left(\frac{1}{t^2} y \right) = \int \frac{te^t}{t^2}$$

$$\Rightarrow \frac{1}{t^2} y = te^t - e^t + c$$

$$\therefore y = t^2 (te^t - e^t + c)$$

Find the solution of Initial condition:-

$$(7) \frac{dy}{dt} = -\frac{y}{1+t} + 2; \quad y(0) = 3$$

$$\Rightarrow \frac{dy}{dt} + \frac{y}{1+t} = 2 \quad \text{--- (i)}$$

$$\begin{aligned} \mu(t) &= e^{\int \frac{1}{1+t} \cdot dt} \\ &= e^{\ln(1+t)} \\ &= (1+t) \end{aligned}$$

multiply the eq (i) by $\mu(t)$ $\Rightarrow$

$$(1+t) \cdot \frac{dy}{dt} + \frac{y(1+t)}{(1+t)} = 2(1+t)$$

$$\therefore \frac{d}{dt} \{ (1+t)y \} = 2(1+t)$$

$$\therefore \int d(1+t)y = \int 2(1+t) \cdot dt$$

$$(1+t)y = 2t + \frac{1}{2} \times 2t^2 + c \quad \text{--- (ii)}$$

$$\therefore y = \frac{1}{1+t} (2t + t^2 + c) \quad \text{--- (iii)}$$

Applying initial condition $y(0) = 3$ where $t = 0, y = 3$
in eq (ii) $\Rightarrow$

$$\therefore (1+0)3 = c$$

$$\therefore c = 3$$

Putting the value of c in eq (iii) $\Rightarrow$

$$y = \frac{1}{1+t} (2t + t^2 + 3)$$

$$(8) \frac{dy}{dt} = \frac{1}{t+1} y + t^2 + t, \quad y(1) = 10$$

$$\Rightarrow \frac{dy}{dt} - \frac{1}{t+1} y = t^2 + t \quad \text{--- (1)}$$

$$\begin{aligned} \therefore I(t) &= e^{-\int \frac{1}{t+1} dt} \\ &= e^{-\ln(t)} \\ &= e^{\ln(t)^{-1}} \\ &= \frac{1}{t} \end{aligned}$$

Multiply the eq(1) by $I(t)$ $\Rightarrow$

$$\frac{1}{t} \cdot \frac{dy}{dt} - \frac{1}{t} \times \left(\frac{1}{t+1} \right) y = \frac{1}{t} (t^2 + t)$$

$$\Rightarrow \frac{d}{dt} \left\{ \frac{1}{t} \cdot y \right\} = t + 1$$

$$\Rightarrow \int d \left(\frac{1}{t} \cdot y \right) = \int (t + 1) dt$$

$$\Rightarrow \frac{1}{t} \cdot y = 4 \times \frac{1}{2} t^2 + t + C$$

$$\Rightarrow \frac{1}{t} y = 2t^2 + t + C \quad \text{--- (11)}$$

Applying initial condition $y(1) = 10$ where $t = 1, y = 10$ in eq(11) $\Rightarrow$

$$\therefore 10 = 2 + 4 + C$$

$$\therefore C = 4$$

Putting the value of C in eq (11) $\Rightarrow$

$$y = t(2t^2 + t + 4)$$

$$\textcircled{1} \quad \frac{dy}{dt} = -\frac{y}{t} + 2 \quad y(1) = 3$$

$$\Rightarrow \frac{dy}{dt} + \frac{y}{t} = 2 \quad \text{--- (1)}$$

$$\begin{aligned} \Rightarrow I(t) &= e^{\int \frac{1}{t} \cdot dt} \\ &= e^{\ln(t)} \\ &= t \end{aligned}$$

Multiply eq (1) by $I(t) \Rightarrow$

$$t \cdot \frac{dy}{dt} + t \cdot \frac{y}{t} = 2t$$

$$\Rightarrow \frac{d}{dt}(t \cdot y) = 2t$$

$$\Rightarrow \int d(t \cdot y) = \int 2t \cdot dt$$

$$\Rightarrow t \cdot y = t^2 + c \quad \text{--- (2)}$$

Applying initial condition: $y(1) = 3$, where, $y = 3$, $t = 1$

$$3 = 1 + c$$

$$\therefore c = 2$$

Putting the value of c in eq (2) $\Rightarrow$

$$y = \frac{1}{t}(t^2 + 2)$$

Ans. -

$$(10) \quad \frac{dy}{dt} = -2ty + 4e^{-t^2}; \quad y(0) = 3$$

$$\Rightarrow \frac{dy}{dt} + 2ty = 4e^{-t^2} \quad \text{--- (1)}$$

$$\Rightarrow I.I. = e^{\int 2t \cdot dt}$$

$$= e^{t^2}$$

multiply the value of I.I. in eq (1):

$$\Rightarrow e^{t^2} \frac{dy}{dt} + e^{t^2} \cdot 2ty = e^{t^2} \cdot 4e^{-t^2}$$

$$\Rightarrow \frac{d}{dt}(e^{t^2} y) = 4e^{t^2} \cdot e^{-t^2}$$

$$\Rightarrow \int d(e^{t^2} y) = \int 4e^{t^2-t^2} \cdot dt$$

$$\Rightarrow e^{t^2} y = 4t + c \quad \text{--- (2)}$$

Applying initial condition $y(0) = 3$ in eq (2):
 where, $t = 0, y = 3$

$$\therefore 3 = c$$

put the value of c in eq (2):

$$e^{t^2} y = 4t + 3$$

$$(71) \quad \frac{dy}{dt} - \frac{2y}{t} = 2t^2 \quad \text{--- (i)}; \quad y(-2) = 4$$

$$\Rightarrow ll(t) = e^{-\int \frac{2}{t} dt}$$

$$= e^{-2 \ln t}$$

$$= e^{\ln(t)^{-2}}$$

$$= \frac{1}{t^2}$$

multiply eq (i) by $ll(t) \Rightarrow$

$$\Rightarrow \frac{1}{t^2} \frac{dy}{dt} - \frac{1}{t^2} \times \frac{2y}{t} = \frac{1}{t^2} 2t^2$$

$$\Rightarrow \frac{d}{dt} \left(\frac{1}{t^2} y \right) = 2$$

$$\Rightarrow \int d \left(\frac{1}{t^2} y \right) = \int 2 dt$$

$$\Rightarrow \frac{1}{t^2} y = 2(t) + c \quad \text{--- (ii)}$$

Applying initial condition $y(-2) = 4$ where $y=4, t=-2$

$$\frac{1}{(-2)^2} \times 4 = 2(-2) + c$$

$$\therefore c = 2 + 4 = 6$$

Put the value of c in eq (ii) $\Rightarrow$

$$y = t^2(2t + 6)$$

Ans:—

$$(22) \quad \frac{dy}{dt} - \frac{3}{t}y = 2t^3 e^{2t}; \quad y(1) = 0 \quad (1)$$

$$\Rightarrow u(t) = e^{-\int \frac{3}{t} \cdot dt}$$

$$\begin{aligned} \Rightarrow &= e^{-3 \ln(t)} \\ &= e^{\ln(t)^{-3}} \\ &= \frac{1}{t^3} \end{aligned}$$

multiply the eq(1) by $u(t)$

$$\frac{1}{t^3} \frac{dy}{dt} - \frac{1}{t^3} \times \frac{3}{t} y = \frac{2t^3 e^{2t}}{t^3}$$

$$\Rightarrow \frac{d}{dt} \left\{ \frac{1}{t^3} \times y \right\} = 2e^{2t}$$

$$\Rightarrow \int d \left(\frac{1}{t^3} \times y \right) = \int 2e^{2t} \cdot dt$$

$$\Rightarrow \frac{1}{t^3} y = 2 \times \frac{1}{2} e^{2t} + C$$

$$\Rightarrow \frac{1}{t^3} y = e^{2t} + C \quad (11)$$

Applying the initial condition $y(1) = 0$ where,
 $y = 0, t = 1$

$$\therefore C = -e^{2 \times 1}$$

Now, put the value of C in eq (11)

$$y = t^3 (e^{2t} - 2)$$

Ans:-

Find the general solution of —

$$(21) \quad \frac{dy}{dt} = 3 - 2y$$

$$\Rightarrow \frac{dy}{3-2y} = dt$$

$$\Rightarrow \int \frac{dy}{3-2y} = \int dt$$

$$\Rightarrow -\frac{1}{2} \ln(3-2y) = t + c$$

$$\Rightarrow \ln(3-2y)^{-\frac{1}{2}} = t + c$$

$$\Rightarrow e^{\ln(3-2y)^{-\frac{1}{2}}} = e^{t+c}$$

$$\Rightarrow (3-2y)^{-\frac{1}{2}} = e^t \cdot e^c$$

$$\Rightarrow \frac{2}{3-2y} = e^t \cdot A$$

$$\Rightarrow 3-2y = \frac{2}{Ae^t}$$

$$\Rightarrow 2y = 3 - \frac{2}{Ae^t}$$

$$\therefore y = \frac{1}{2} \left(3 - \frac{2}{Ae^t} \right)$$

Ans:-

$$(22) \quad \frac{dy}{dt} = ty$$

$$\Rightarrow \frac{dy}{y} = t dt$$

$$\Rightarrow \int \frac{dy}{y} = \int t \cdot dt$$

$$\Rightarrow \ln(y) = \frac{1}{2} t^2 + c$$

$$\Rightarrow e^{\ln(y)} = e^{\frac{1}{2} t^2 + c}$$

$$\Rightarrow y = e^{\frac{1}{2} t^2} \cdot A$$

Ans:-

$$(23) \quad \frac{dy}{dt} = 3y + e^{7t}$$

$$\Rightarrow \frac{dy}{dt} - 3y = e^{7t} \quad \text{--- (1)}$$

$$I(t) = e^{\int -3 dt}$$

$$= e^{-3t}$$

Multiply eq (1) by $I(t) \Rightarrow$

$$e^{-3t} \cdot \frac{dy}{dt} - 3y \cdot e^{-3t} = e^{-3t} \cdot e^{7t}$$

$$\frac{d}{dt} (e^{-3t} \cdot y) = e^{7t-3t}$$

$$\int d(e^{-3t} \cdot y) = \int e^{4t} \cdot dt$$

$$e^{-3t} \cdot y = \frac{1}{4} e^{4t}$$

$$(24) \quad \frac{dy}{dt} = \frac{y}{1+t^2}$$

$$\Rightarrow \frac{dy}{y} = \frac{t}{1+t^2} \cdot dt$$

$$\Rightarrow \int \frac{dy}{y} = \int \frac{t}{1+t^2} \cdot dt$$

$$\Rightarrow \ln(y) = \frac{1}{2} \ln(1+t^2) + c$$

$$\Rightarrow \ln(y) - \ln(1+t^2)^{\frac{1}{2}} = c$$

$$\Rightarrow \ln\left(\frac{y}{(1+t^2)^{\frac{1}{2}}}\right) = c$$

$$\Rightarrow e^{\ln\left(\frac{y}{(1+t^2)^{\frac{1}{2}}}\right)} = e^c$$

$$\Rightarrow \frac{y}{(1+t^2)^{\frac{1}{2}}} = A$$

$$\therefore y = A(1+t^2)^{\frac{1}{2}}$$

$$(25) \quad \frac{dy}{dt} = -5y + \sin 3t$$

$$\Rightarrow \frac{dy}{dt} + 5y = \sin 3t \quad \text{--- (1)}$$

$$\Rightarrow \mu(t) = e^{\int 5 dt}$$

$$= e^{5t}$$

multiply eq (1) by $\mu(t) \Rightarrow$

$$e^{5t} \frac{dy}{dt} + e^{5t} \cdot 5y = e^{5t} \sin 3t$$

$$\Rightarrow \int d(e^{5t} \cdot y) = \int e^{5t} \cdot \sin 3t \, dt$$

$$\Rightarrow e^{5t} \cdot y = \frac{e^{5t}(5 \sin 3t - 3 \cos 3t)}{5^2 + 3^2} + C$$

$$\Rightarrow y = \frac{5 \sin 3t - 3 \cos 3t}{34} + C$$

Ans:-

Formula:-

$$\textcircled{1} \int e^{ax} \sin bx \, dx = \frac{e^{ax}(a \sin bx - b \cos bx)}{a^2 + b^2} + C$$

$$\textcircled{2} \int e^{ax} \cos bx \, dx = \frac{e^{ax}(a \cos bx + b \sin bx)}{a^2 + b^2} + C$$

$$\textcircled{26} \frac{dy}{dt} = t + \frac{2y}{1+t}$$

$$\Rightarrow \frac{dy}{dt} - \frac{2y}{1+t} = t \quad \text{--- (1)}$$

$$\begin{aligned} \therefore I(t) &= \int \frac{-2}{1+t} \cdot dt \\ &= -2 \ln(1+t) \\ &= e \end{aligned}$$

Multiply the eq (1) by $I(t) \Rightarrow$

$$(1+t)^{-2} \frac{dy}{dt} - (1+t)^{-2} \cdot \frac{2y}{1+t} = t(1+t)^{-2}$$

$$\Rightarrow \frac{d}{dt} \left((1+t)^{-2} \cdot y \right) = t \cdot \frac{1}{(1+t)^2}$$

$$\Rightarrow \int d \left\{ \frac{1}{(1+t)^2} y \right\} = \int \frac{t}{(1+t)^2} \cdot dt$$

$$\Rightarrow \frac{1}{(1+t)^2} y = \ln(1-t) + \frac{1}{1-t} + C$$

$$\therefore y = (1-t)^2 \left\{ \ln(1-t) + \frac{1}{1-t} + C \right\}$$

$$(27) \quad \frac{dy}{dt} = 3 + y^2$$

$$\Rightarrow \frac{dy}{3+y^2} = dt$$

$$\Rightarrow \int \frac{dy}{3+y^2} = \int dt$$

$$\Rightarrow \int \frac{dy}{(\sqrt{3})^2 + y^2} = \int dt$$

$$\Rightarrow \frac{1}{\sqrt{3}} + \tan^{-1}\left(\frac{y}{\sqrt{3}}\right) = t + c$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{\sqrt{3}}\right) = \sqrt{3}(t + c)$$

$$\Rightarrow \tan^{-1}\left(\frac{y}{\sqrt{3}}\right) = \sqrt{3}(t + c)$$

$$\Rightarrow \frac{y}{\sqrt{3}} = \tan(\sqrt{3}(t + c))$$

$$\therefore y = \sqrt{3} \tan\{\sqrt{3}(t + c)\}$$

Ans:-

$$(20) \frac{dy}{dt} = -3y + e^{-2t} + t^2$$

$$\Rightarrow \frac{dy}{dt} + 3y = e^{-2t} + t^2 \quad \text{--- (1)}$$

$$\therefore \mu(t) = e^{\int 3 dt}$$

$$= e^{3t}$$

now multiply (1) by $\mu(t)$

$$e^{3t} \frac{dy}{dt} + e^{3t} \cdot 3y = e^{-2t} \cdot e^{3t} + e^{3t} \cdot t^2$$

$$\frac{dy}{dt} \{e^{3t} \cdot y\} = e^{-2t+3t} + t^2 e^{3t}$$

$$\int d(e^{3t} \cdot y) = \int (e^t + \frac{t^2}{\frac{1}{e^{3t}}}) dt$$

$$e^{3t} y = e^t + \frac{1}{3} t^2 e^{3t} + \frac{1}{9} t e^{3t} + \frac{1}{27} e^{3t} + C$$

$$\therefore e^{3t} y = e^t + \frac{1}{3} t^2 e^{3t} + \frac{1}{9} t e^{3t} + \frac{1}{27} e^{3t} + C$$

Higher Order
Linear Diff. Equations!

Higher Order linear Differential equation:—

⇒ There are two types of Higher Order linear Differential equation.

- ① Non Homogeneous
- ② Homogeneous.

✎ Non-Homogeneous n -th Order linear D.E:—

$$\textcircled{1} \quad a_1 \frac{d^ny}{dx^n} + a_2 \frac{d^{n-1}y}{dx^{n-1}} + a_3 \frac{d^{n-2}y}{dx^{n-2}} + \dots + a_n y = f(x)$$

where—

$$\begin{aligned} f(x) &= e^{\alpha x} \\ &= \sin x \\ &= \sin x + \cos x \\ &= x^2 + 2 \\ &= x^2 + 2x + 3 \end{aligned}$$

✎ Homogeneous n -th Order linear D.E:—

$$\textcircled{2} \quad a_1 \frac{d^ny}{dx^n} + a_2 \frac{d^{n-1}y}{dx^{n-1}} + a_3 \frac{d^{n-2}y}{dx^{n-2}} + \dots + a_n y = 0$$

Find the general solution of the
Following DE: —————

① $\frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 10y = 0$ or

$a_1 \frac{d^2y}{dt^2} + a_2 \frac{dy}{dt} + a_3 y = 0$

where, $a_1, a_2, a_3 = \text{constant}$

Formula: —————

* A trial solution $y = e^{mt}$ leads to the Auxiliary equation (AE)

$a_1 m^2 + a_2 m + a_3 = 0$ or

$m = \frac{-a_2 \pm \sqrt{a_2^2 - 4a_1a_3}}{2a_1}$

Condition: —————

① If roots are real and Unique.

Ex: $m = 2, 3$

$\therefore y = c_1 e^{2x} + c_2 e^{3x}$

② If roots are real and equal:—

Ex: $m=2, 2$

$$\therefore y = (c_1 + c_2 x) e^{2x}$$

③ If roots are complex and conjugate: $(a \pm ib)$:—

Ex: $m = 2 \pm i3$

$$\therefore y = e^{2x} (c_1 \cos 3x + c_2 \sin 3x)$$

Find the general solution of
the following D.E:—

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① $\frac{d^2 y}{dt^2} + 7 \frac{dy}{dt} + 10y = 0$

$\Rightarrow$ Let, $y = e^{mt}$ be a trial solution of the equation

$$\therefore \frac{d^2 y}{dt^2} = m^2 e^{mt} ; \frac{dy}{dt} = m e^{mt}$$

So, the equation is:—

$$m^2 e^{mt} + 7m e^{mt} + 10e^{mt} = 0$$

$$\Rightarrow e^{mt} (m^2 + 7m + 10) = 0$$

$$\Rightarrow m^2 + 7m + 10 = 0 \quad [e^{mt} \neq 0]$$

Now, The Auxiliary equation is:—

$$m^2 + 7m + 10 = 0$$

$$m^2 + 5m + 2m + 10 = 0$$

$$m(m+5) + 2(m+5) = 0$$

$$(m+2)(m+5) = 0$$

$$m = -2, -5$$

The roots are real and Unique:—

$\therefore$ The general Solution is:—

$$y = C_1 e^{-5t} + C_2 e^{-2t}$$

$$\textcircled{2} \quad \frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y = 0$$

Let $y = e^{mt}$ be a trial solution of the equation:—

So, The Auxiliary equation is:—

$$m^2 + 5m + 6 = 0$$

$$\therefore m = -2, -3$$

The roots are real and Unique:—

∴ The general Solution is:—

$$y = c_1 e^{-3t} + c_2 e^{-2t}$$

③ $\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + y = 0$

Let, $y = e^{mt}$ be a trial Solution of the equation.

So, The Auxiliary equation is:—

$$m^2 + 4m + 1 = 0$$

$$m = \frac{-4 \pm \sqrt{(4)^2 - 4 \times 1 \times 1}}{2 \times 1}$$

$$= -2 \pm \sqrt{3}$$

$$\therefore m = -2 - \sqrt{3}, -2 + \sqrt{3}$$

The roots are real and Unique.

∴ The general Solution is:—

$$\begin{aligned} y &= c_1 e^{(-2-\sqrt{3})t} + c_2 e^{(-2+\sqrt{3})t} \\ &= c_1 e^{-3.73t} + c_2 e^{-0.26t} \end{aligned}$$

$$(4) \frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 7y = 0$$

Let, $y = e^{mt}$ be a trial solution of the equation.

So, The Auxiliary equation is:—

$$m^2 + 6m + 7 = 0$$

$$\therefore m = \frac{-6 \pm \sqrt{6^2 - 4 \times 7}}{2 \times 1}$$

$$\therefore m = -3 + \sqrt{2}, -3 - \sqrt{2}$$

$$= -1.585, -4.414$$

The roots are real and Unique.

$\therefore$ The general Solution is:—

$$y = e_1 e^{(-3-\sqrt{2})t} + e_2 e^{(-3+\sqrt{2})t}$$

$$= e_1 e^{-4.41t} + e_2 e^{-1.585t}$$

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$$(1) \frac{d^2y}{dt^2} - 6 \frac{dy}{dt} - 7y = 0$$

Let $y = e^{mt}$ be a trial solution of the equation.

So, The Auxiliary equation is:—

$$m^2 - 6m - 7 = 0$$

$$\Rightarrow m^2 - 7m + m - 7 = 0$$

$$\Rightarrow m(m-7) + 1(m-7) = 0$$

$$\Rightarrow (m+1)(m-7) = 0$$

$$\Rightarrow m = 7, -1$$

The roots are real and Unique.

$\therefore$ The general solution is:—

$$y = c_1 e^{-t} + c_2 e^{7t}$$

$$(2) \frac{d^2y}{dt^2} - \frac{dy}{dt} - 12y = 0$$

Let $y = e^{mt}$ be a trial solution of the equation.

So, The Auxiliary equation is:—

$$m^2 - m - 12 = 0$$

$$\therefore m = 4, -3.$$

The roots are real and Unique.

So, The general Solution is:

$$y = c_1 e^{-3t} + c_2 e^{4t}$$

$$\textcircled{2} \quad \frac{d^2 y}{dt^2} + 6 \frac{dy}{dt} + 9y = 0$$

Let, $y = e^{mt}$ be a trial solution of the equation:-

So, The Auxiliary equation is:

$$m^2 + 6m + 9 = 0$$

$$\Rightarrow m^2 + 3m + 3m + 9 = 0$$

$$\Rightarrow m(m+3) + 3(m+3) = 0$$

$$\Rightarrow (m+3)(m+3) = 0$$

$$\therefore m = -3, -3$$

The roots are real and equal.

$\therefore$ The general Solution is:

$$y = (c_1 + c_2 t) e^{-3t}$$

$$(4) \frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 4y = 0$$

Let, $y = e^{mt}$ be a trial solution of the equation

So, The Auxiliary equation:—

$$m^2 - 4m + 4 = 0$$

$$\Rightarrow m^2 - 2m - 2m + 4 = 0$$

$$\Rightarrow m(m-2) - 2(m-2) = 0$$

$$\Rightarrow (m-2)(m-2) = 0$$

$$\therefore m = 2, 2$$

The roots are real and equal.

$\therefore$ The General Solution is:—

$$y = (c_1 + c_2 t) e^{2t}$$

$$(5) \frac{d^2y}{dt^2} + 8\frac{dy}{dt} + 25y = 0$$

Let, $y = e^{mt}$ be a trial solution of the equation.

So, The Auxiliary equation:—

$$m^2 + 8m + 25 = 0$$

$$m = \frac{-8 \pm \sqrt{8^2 - 4 \times 25}}{2}$$

$$= \frac{-8 \pm \sqrt{-36}}{2}$$

$$\therefore m = \frac{-8 \pm i6}{2}$$

$$= -4 \pm i3$$

$$m = -4 + i3, -4 - i3$$

The roots are Complex and Conjugate

$\therefore$ The general Solution is:—

$$y = e^{-4t} (c_1 \cos 3t + c_2 \sin 3t)$$

$$\textcircled{6} \quad \frac{d^2 y}{dt^2} - 4 \frac{dy}{dt} + 20y = 0$$

Let $y = e^{mt}$ be a trial Solution of the equation.

So, The Auxiliary equation is:—

$$m^2 - 4m + 20 = 0$$

$$m = \frac{4 \pm \sqrt{4^2 - 4 \times 20}}{2}$$

$$= \frac{4 \pm \sqrt{-100}}{2}$$

$$= \frac{4 \pm 10i}{2}$$

$$= 2 \pm 5i$$

The roots are complex and conjugate.

∴ The general Solution is:—

$$y = e^{2t} (c_1 \cos t + c_2 \sin t)$$

Find the solution of the
given initial value problem:-

For Homogeneous:-

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7. $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = 0$; $y(0) = 6, y'(0) = -2$ — (i)

Let, $y = e^{mt}$ be a trial solution of the eq - (i)

So, Auxiliary equation:-

$$m^2 + 2m - 3 = 0$$

$$\therefore m = -3, 1$$

roots are real and unequal.

The general solution is:-

$$y = c_1 e^{-3t} + c_2 e^t \quad \text{--- (ii)}$$

$$y' = -3c_1 e^{-3t} + c_2 e^t \quad \text{--- (iii)}$$

using $y(0) = 6$ in eq (ii) $\Rightarrow$

$$6 = c_1 e^{-3 \times 0} + c_2 e^0$$

$$6 = c_1 + c_2 \quad \text{--- (iv)}$$

using $y'(0) = -2$ in eq (iii) $\Rightarrow$

$$-2 = -3c_1 e^0 + c_2 e^0$$

$$\text{--- (v)}$$

$$-2 = -3c_1 + c_2$$

$$\textcircled{IV} - \textcircled{V} =$$

$$e_1 + e_2 = 6$$

$$-3e_1 + e_2 = -2$$

$$\begin{array}{r} (+) \quad (-) \quad (+) \\ \hline \end{array}$$

$$4e_1 = 8$$

$$\therefore e_1 = 2$$

Put the value of e_1 in eq $\textcircled{IV}$:-

$$e_2 = 6 - 2 = 4$$

So, the general solution is:-

$$y = 2e^{-3t} + 4e^t$$

Ans:-

$$\textcircled{8} \quad \frac{d^2y}{dt^2} + 4 \frac{dy}{dt} - 5y = 0 \quad y(0) = 1, y'(0) = -7 \quad \textcircled{1}$$

Let, $y = e^{mt}$ be a trial solution of eq $\textcircled{1}$:-

So Auxiliary Equation:-

$$m^2 + 4m - 5 = 0$$

$$\therefore m = -5, 1$$

Roots are real and unequal.

The general solution is:-

$$y = e_1 e^{-5t} + e_2 e^t \quad \textcircled{II}$$

$$y' = -5e_1 e^{-5t} + e_2 e^t \quad \textcircled{III}$$

Using $y(0) = 11$ in eq (II) $\Rightarrow$

$$c_1 + c_2 = 11 \quad \text{--- (IV)}$$

Using $y'(0) = 7$ in eq (III) $\Rightarrow$

$$-5c_1 + c_2 = -7 \quad \text{--- (V)}$$

(IV) + (V) $\Rightarrow$

$$c_1 + c_2 = 11$$

$$-5c_1 + c_2 = -7$$

$$\begin{array}{r} (+) \quad (+) \quad (+) \\ \hline \end{array}$$

$$6c_1 = 18$$

$$\therefore c_1 = \frac{18}{6} = 6/2 = 3$$

Put the value of c_1 in eq (IV) $\Rightarrow$

$$c_2 = 11 - 3 = 8$$

The general solution:-

$$\therefore y = 3e^{-5t} + 8e^t$$

Ans:-

$$(11) \quad \frac{d^2y}{dt^2} - 8\frac{dy}{dt} + 16y = 0 \quad y(0) = 3, \quad y'(0) = 11 \quad \text{--- (1)}$$

Let, $y = e^{mt}$ be a trial solution of the eq (1) =

So, The Auxiliary equation is:—

$$m^2 - 8m + 16 = 0$$

$$m = 4, 4$$

roots are real and equal.

The general solution:—

$$y = (c_1 + c_2 t)e^{4t} \quad \text{--- (ii)}$$

$$y' = 4(c_1 + c_2 t)e^{4t} \quad \text{--- (iii)}$$

$$\Rightarrow y' = 4c_1 e^{4t} + 4c_2 t e^{4t} + c_2 e^{4t} \quad \text{--- (iii)}$$

Using $y(0) = 3$ in eq (ii) =

$$c_1 = 3$$

Using $y'(0) = 11$ in eq (iii) =

$$c_2 = -1$$

From eq (ii) =

$$y = (3 - t)e^{4t} \\ = 3e^{4t} - te^{4t}$$

Ans:—

12) $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 4y = 0, \quad y(0) = 1, \quad y'(0) = 1 \quad \text{--- (i)}$

Let, $y = e^{mt}$ be a trial solution of the eq (i) =

So, The Auxiliary equation is:

$$m^2 - 4m + 4 = 0$$

$$\therefore m = 2, 2$$

Roots are real and equal.

The general solution:

$$y = (c_1 + c_2 t) e^{2t} \quad \text{--- (ii)}$$

$$y' = 2(c_1 + c_2 t) e^{2t} \quad \text{--- (iii)} \Rightarrow 2c_1 e^{2t} + 2c_2 t e^{2t} + c_2 e^{2t} \quad \text{--- (iii)}$$

using $y(0) = 1$ in eq (ii) =

$$c_1 = 1$$

using $y'(0) = 1$ in eq (iii) =

$$c_2 = -1$$

From eq (ii) =

$$y = (1 - t) e^{2t} \\ = e^{2t} - t e^{2t}$$

Ans.

Non-Homogeneous

LDF



#Formula:-

* $L(D)y = f(x)$

* The general solution $y = y_c + y_p$

where,

y_c = general solution $L(D)y = 0$ and

y_p = Particular solution of $L(D)y = f(x)$

* y_c = Complementary function (CF)

y_p = Particular Integral (PI)

The Method of Undetermined Co-efficient

$L(D)y = a_2 \frac{d^2y}{dx^2} + a_1 \frac{dy}{dx} + a_0 = F(x)$

$F(x)$	y_p (Particular Integral) (PI)
$k = \text{Any Constant}$	A
$ax+b / ax$	$Ax+B$
ax^2+Bx+C	Ax^2+Bx+C
ae^{mx}	Ae^{mx}
$a \sin x / b \cos x$	$A \cos x + B \sin x$
$a \sin x + b \cos x$	$A \cos x + B \sin x$

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Find the general solution of the following D.E.:-

$$(1) \quad \frac{d^2y}{dt^2} - \frac{dy}{dt} - 6y = e^{4t} \quad \text{--- (1)}$$

Consider:-

$$\frac{d^2y}{dt^2} - \frac{dy}{dt} - 6y = 0 \quad \text{--- (2)}$$

Let $y = e^{mt}$ be a trial solution of the equation (2)

So, The Auxiliary equation:-

$$m^2 - m - 6 = 0$$

$$\therefore m = -2, 3$$

Roots are real and unequal.

$\therefore$ The Complementary function:-

$$y_c = c_1 e^{-2t} + c_2 e^{3t}$$

Now The particular Integral:-

$$y_p = Ae^{4t}$$

$$y_p' = 4Ae^{4t}$$

$$y_p'' = 16Ae^{4t}$$

putting the value of y_p, y_p' in eq (1) $\Rightarrow$

$$\Rightarrow 16Ae^{4t} - 4Ae^{4t} - 6Ae^{4t} = e^{4t}$$

$$\Rightarrow 6Ae^{4t} = e^{4t}$$

$$\therefore A = \frac{1}{6}$$

$\therefore$ The particular Integral:

$$y_p = \frac{1}{6} e^{4t}$$

$\therefore$ The general solution:

$$\begin{aligned} y &= y_c + y_p \\ &= c_1 e^{-2t} + c_2 e^{3t} + \frac{1}{6} e^{4t} \end{aligned}$$

$$\textcircled{2} \quad \frac{d^2 y}{dt^2} + 6 \frac{dy}{dt} + 8y = 2e^{-3t} \quad \text{--- (1)}$$

Consider,

$$\frac{d^2 y}{dt^2} + 6 \frac{dy}{dt} + 8y = 0 \quad \text{--- (ii)}$$

Let, $y = e^{mt}$ be a trial solution of the equation (ii) $\Rightarrow$

$\therefore$ The Auxiliary Equation:

$$m^2 + 6m + 8 = 0$$

$$\therefore m = -2, -4$$

The roots are real and unequal.

The Complementary function:-

$$Y_c = e_1 e^{-4t} + e_2 e^{-2t}$$

Now- The particular Integral:-

$$Y_p = A e^{-3t}$$

$$Y_p' = -3A e^{-3t}$$

$$Y_p'' = +9A e^{-3t}$$

Putting the Y_p' , Y_p'' in eq (1) :-

$$9A e^{-3t} - 18A e^{-3t} + 8A e^{-3t} = 2e^{-3t}$$

$$17A e^{-3t} - 18A e^{-3t} = 2e^{-3t}$$

$$-A e^{-3t} = 2e^{-3t}$$

Equating like terms:-

$$-A = 2$$

$$\therefore A = -2$$

$\therefore$ The Particular Integral:-

$$Y_p = -2e^{-3t}$$

$\therefore$ The general Equation:-

$$Y = Y_p + Y_c$$

$$= e_1 e^{-4t} + e_2 e^{-2t} - 2e^{-3t}$$

$$\textcircled{3} \quad \frac{d^2y}{dt^2} - \frac{dy}{dt} - 2y = 5e^{3t} \quad \text{--- (I)}$$

Consider:-

$$\frac{d^2y}{dt^2} - \frac{dy}{dt} - 2y = 0 \quad \text{--- (II)}$$

Let, $y = e^{mt}$ be a trial solution of the equation (II) $\Rightarrow$

So, The Auxiliary equation:-

$$m^2 - m - 2 = 0$$

$$\therefore m = \frac{1 \pm \sqrt{1+8}}{2}$$

Roots are complex conjugate.

$\therefore$ The Complementary function:-

$$y_c = e^{\frac{1}{2}t} \left(c_1 \cos \frac{\sqrt{7}}{2}t + c_2 \sin \frac{\sqrt{7}}{2}t \right)$$

Now:- The Particular Integral.

$$y_p = Ae^{3t}$$

$$y_p' = 3Ae^{3t}$$

$$y_p'' = 9Ae^{3t}$$

Putting the value of y_p, y_p', y_p'' in eq (I) $\Rightarrow$

$$9Ae^{3t} - 3Ae^{3t} - 2Ae^{3t} = 5e^{3t}$$

$$\therefore A = \frac{1}{4} \times 5 = \frac{5}{4}$$

∴ The particular Integral:-

$$Y_p = \frac{8}{4} e^{3t}$$

∴ The general Solution is:-

$$Y = Y_e + Y_p$$

$$= e^{\frac{1}{2}t} \left(c_1 \cos \frac{\sqrt{3}}{2}t + c_2 \sin \frac{\sqrt{3}}{2}t \right) + \frac{8}{4} e^{3t}$$

Ans:-

Q. $\frac{d^2y}{dt^2} + 4 \frac{dy}{dt} + 13y = e^{-t}$ ——— (1)

Consider:-

$$\frac{d^2y}{dt^2} + 4 \frac{dy}{dt} + 13y = 0 \quad \text{————— (11)}$$

Let: $y = e^{mt}$ be a trial solution of the eq (1) ⇒

so, The Auxiliary equation:-

$$m^2 + 4m + 13 = 0$$

$$\therefore m = -2 \pm 3i$$

The roots are complex conjugate.

The Complementary function:-

$$Y_c = e^{-2t} (c_1 \cos 3t + c_2 \sin 3t)$$

Now, The Particular Integral: (PI)

$$y_p = Ae^{-t}$$

$$y_p' = -Ae^{-t}$$

$$y_p'' = Ae^{-t}$$

Putting the all value of y_p in eq. (1)

$$Ae^{-t} - 4Ae^{-t} + 10Ae^{-t} = e^{-t}$$

$$10Ae^{-t} = e^{-t}$$

$$\therefore A = \frac{1}{10}$$

$\therefore$ PI

$$y_p = \frac{1}{10}e^{-t}$$

$\therefore$ The general Solution:

$$y = y_c + y_p$$

$$= e^{-2t}(c_1 \cos t + c_2 \sin t) + \frac{1}{10}e^{-t}$$

Ans.

$$(5) \frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 13y = -3e^{-2t} \quad \text{--- (1)}$$

Consider ---
mon

$$\frac{d^2 y}{dt^2} + 4 \frac{dy}{dt} + 13y = 0 \quad \text{--- (11)}$$

Let, $y = e^{mt}$ be a trial solution of the eq (11) $\Rightarrow$

So, Auxiliary equation: ---
mon

$$m^2 + 4m + 13 = 0$$

$$\therefore m = -2 \pm 3i$$

the roots are complex conjugate.

$\therefore$ Complementary function: ---
mon

$$y_c = e^{-2t} (c_1 \cos 3t + c_2 \sin 3t)$$

Now: --- Particular Integral: ---
mon

$$y_p = Ae^{-2t}$$

$$y_p' = -2Ae^{-2t}$$

$$y_p'' = 4Ae^{-2t}$$

From eq (1) $\Rightarrow$
mon

$$4Ae^{-2t} - 8Ae^{-2t} + 13Ae^{-2t} = -3e^{-2t}$$

$$\therefore A = -\frac{1}{3}$$

∴ The particular Integral:—
w o o o o

$$Y_p = -\frac{1}{9} e^{-2t}.$$

∴ The general Solution:—
w o o o o

$$Y = Y_c + Y_p$$

$$= e^{-2t} (c_1 \cos t + c_2 \sin t) - \frac{1}{9} e^{-2t}.$$

Ans:—

$$(6) \quad \frac{d^2 y}{dt^2} + 7 \frac{dy}{dt} + 10y = e^{-2t} \quad \text{--- (I)}$$

Consider:

$$\frac{d^2 y}{dt^2} + 7 \frac{dy}{dt} + 10y = 0 \quad \text{--- (II)}$$

Let, $y = e^{mt}$ be a trial solution of the eq (II) ⇒

So, The Auxiliary equation:—
w o o o o

$$m^2 + 7m + 10 = 0$$

$$\therefore m = -5, -2$$

The roots are real and unequal:—

∴ The Complimentary function:—
w o o o o

$$Y_c = c_1 e^{-5t} + c_2 e^{-2t}$$

∴ The Particular Integral:—
monom

$$y_p = Ae^{-2t}$$

$$y_p' = -2Ae^{-2t}$$

$$y_p'' = 4Ae^{-2t}$$

Forcom eq (1) ⇒
monom

$$4Ae^{-2t} - 14Ae^{-2t} + 10Ae^{-2t} = e^{-2t}$$

∴ The PI:—
monom

$$y_p =$$

∴ The general Solution:—
monom

$$y = y_p + y_c$$

$$= c_1 e^{-t} + c_2 e^{-2t} + y_p$$

Ans:—

$$(7) \frac{d^2y}{dt^2} - 5 \frac{dy}{dt} + 4y = e^{4t} \quad \text{--- (1)}$$

Consider:
monom

$$\frac{d^2y}{dt^2} - 5 \frac{dy}{dt} + 4y = 0 \quad \text{--- (2)}$$

Let, $y = e^{mt}$ be a trial solution of the eq (2) ⇒

so, Auxiliary equation:—
monom

$$m^2 - 5m + 4 = 0$$

$$m = 1, 4$$

The roots are real and unequal—

So, The Complementary function:-

$$y_c = c_1 e^t + c_2 e^{4t}$$

Now, The particular Integral:-

$$y_p = Ae^{4t}$$

$$y_p' = 4Ae^{4t}$$

$$y_p'' = 16Ae^{4t}$$

For: eq (1) :-

$$16Ae^{4t} - 20Ae^{4t} + 4Ae^{4t} = e^{4t}$$

$$16Ae^{4t} - 20Ae^{4t} + 4Ae^{4t} = e^{4t}$$

The PI:-

$$y_p =$$

Ans:-

$$y = y_p + y_c$$

$$= c_1 e^t + c_2 e^{4t} + y_p$$

Q. $\frac{d^2 y}{dt^2} + \frac{dy}{dt} - 6y = 4e^{-2t}$

Complementary function:-

$$y_c = c_1 e^{-3t} + c_2 e^{2t}$$

For Non-Homogeneous:—
~~~~~

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~~~~~

$$\textcircled{i} \quad \frac{d^2y}{dt^2} + 6 \frac{dy}{dt} + 8y = e^{-t} \quad \text{—————} \textcircled{1} \quad y(0) = y'(0) = 0$$

Consider:—
~~~~~

$$\frac{d^2y}{dt^2} + 6 \frac{dy}{dt} + 8y = 0 \quad \text{—————} \textcircled{ii}$$

Let  $y = e^{mt}$  be a trial sol<sup>n</sup> of eq  $\textcircled{ii}$  =

So, The Auxiliary equation:—  
~~~~~

$$m^2 + 6m + 8 = 0$$

$$\therefore m = -4, -2$$

roots are real and Unequal

$\therefore$ Complementary function:—
~~~~~

$$y_c = c_1 e^{-4t} + c_2 e^{-2t}$$

Now, The particular Integral:—  
~~~~~

$$y_p = A e^{-t}$$

$$y_p' = -A e^{-t}$$

$$y_p'' = A e^{-t}$$

From eq $\textcircled{i}$ =
~~~~~

$$A e^{-t} + 6(-A e^{-t}) + 8A e^{-t} = e^{-t}$$

$$A e^{-t} - 6A e^{-t} + 8A e^{-t} = e^{-t}$$

$$\therefore A = \frac{1}{3}$$

∴ The P.I. —

$$Y_p = \frac{1}{3}e^{-t}$$

∴ The General Solution —

$$Y = Y_p + Y_c$$

$$Y = \frac{1}{3}e^{-t} + c_1 e^{-4t} + c_2 e^{-2t} \quad \text{--- (iii)}$$

$$Y' = -\frac{1}{3}e^{-t} - 4c_1 e^{-4t} - 2c_2 e^{-2t} \quad \text{--- (iv)}$$

From (iii) using  $Y(0) = 0$

$$c_1 + c_2 = -\frac{1}{3} \quad \text{--- (v)}$$

From eq. (iv) using  $Y'(0) = 0$

$$4c_1 - 2c_2 = \frac{1}{3} \quad \text{--- (vi)}$$

$$2 \times \text{(v)} + \text{(vi)} =$$

$$2c_1 + 2c_2 = -\frac{2}{3}$$

$$4c_1 - 2c_2 = \frac{1}{3}$$

$$6c_1 = -\frac{2}{3} + \frac{1}{3}$$

$$\therefore c_1 = -\frac{1}{18}$$

Put the value of  $c_1$  in eq. (v) =

$$c_1 + c_2 = -\frac{1}{3}$$

$$\therefore c_2 = -\frac{1}{3} + \frac{1}{18} = -\frac{5}{18}$$

From eq. (iii) =

$$Y = \frac{1}{3}e^{-t} - \frac{1}{18}e^{-4t} - \frac{5}{18}e^{-2t}$$

Ans. —



$$(1) \quad \frac{d^2y}{dt^2} + 4 \frac{dy}{dt} + 13y = -3e^{-2t}; y(0) = y'(0) = 0 \quad \text{--- (1)}$$

Consider:-

$$\frac{d^2y}{dt^2} + 4 \frac{dy}{dt} + 13y = 0 \quad \text{--- (11)}$$

Let,  $y = e^{mt}$  be a trial soln of eqn (11):

So, The AE:—  
non

$$m^2 + 4m + 13 = 0$$

$$\therefore m = -2 \pm 3i$$

roots are complex conjugate.

$\therefore$  The CF:—  
non

$$y_e = e^{-2t} (c_1 \cos 3t + c_2 \sin 3t)$$

Now:- The PI:—  
non

$$y_p = Ae^{-2t}$$

$$y_p' = -2Ae^{-2t}$$

$$y_p'' = 4Ae^{-2t}$$

From eqn (1):  
non

$$4Ae^{-2t} - 12Ae^{-2t} + 13Ae^{-2t} = -3e^{-2t}$$

$$\therefore A = -\frac{3}{10}$$

So, The PI:—  
non

$$y_p = -\frac{3}{10}e^{-2t}$$

The general solution:—  
non

$$y = y_e + y_p = e^{-2t} c_1 \cos 3t + c_2 \sin 3t e^{-2t} - \frac{3}{10}e^{-2t} \quad \text{--- (11)}$$

$$y' = -2e^{-2t} c_1 \cos 3t - 3e^{-2t} c_1 \sin 3t + 3c_2 e^{-2t} \cos 3t - 2e^{-2t} c_2 \sin 3t - \frac{6}{10}e^{-2t} \quad \text{--- (12)}$$

using  $y(0) = 0$  in eq (ii)  $\Rightarrow$

$$C_1 = \frac{3}{10}$$

using  $y(0) = 0$  in eq (v)  $\Rightarrow$

$$-2C_1 + 3C_2 = -\frac{6}{10}$$

$$\therefore 3C_2 = 2C_1 - \frac{6}{10}$$

$$C_2 = \frac{2 \times \frac{3}{10}}{3} - \frac{6}{10}$$

$$= 0$$

From eq (iii)  $\Rightarrow$

$$y = \frac{3}{10} e^{-2t} \cos 2t - \frac{3}{10} e^{-2t}$$

Ans:-

# Compute the general soln'

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(32)  $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} = 3t + 2$  ——— (I)

Consider,

$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} = 0$  ——— (II)

Let,  $y = e^{mt}$  be a trial soln of eq<sup>n</sup> (II)  $\Rightarrow$

So, AE:-

$m^2 + 2m + 0 = 0$

$\therefore m = 0, -2$

roots are real and unequal.

$\therefore$  The CF:-

$y_c = c_1 e^{-2t} + c_2 e^{0t}$

Now, P.I:-

$y_p = A + B$

$y_p' = A$

$y_p'' = 0$

From eq<sup>n</sup> (I)

$2A = 3t + 2$

The Question is not Right



$$(23) \frac{d^2y}{dt^2} + 4y = 2t + 2 \quad \text{--- (1)}$$

Consider —  
hom

$$\frac{d^2y}{dt^2} + 4y = 0 \quad \text{--- (ii)}$$

Let  $y = e^{mt}$  be a trial sol<sup>n</sup> of eq (ii)

so, The Auxiliary eqn is:—

$$m^2 + 4 = 0$$

$$\therefore m = \pm 2i$$

Roots are Complex Conjugate

$\therefore$  The Complementary function:—

$$y_c = (c_1 \cos 2t + c_2 \sin 2t)$$

Now— Particular Integral:—

$$y_p = At + B \quad \text{--- (iii)}$$

$$y_p' = A$$

$$y_p'' = 0$$

From eq (1):

$$4A - 4B = 2t + 2$$

Eqting like terms:—

$$A = 3/4$$

$$B = 2/4 = 1/2$$

Put the value in eq (iii)

$$y_p = \frac{3}{4}t + \frac{1}{2}$$

$\therefore$  The general sol<sup>n</sup> is:-

$$y = y_p + y_c$$

$$y = \frac{3}{4}t + \frac{1}{2} + c_1 \cos 2t + c_2 \sin 2t$$

Ans:-

$$\textcircled{34} \quad \frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = t^2 \quad \textcircled{i}$$

Consider:-

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 0 \quad \textcircled{ii}$$

Let  $y = e^{mt}$  be a trial soln of eq  $\textcircled{ii} \Rightarrow$

So, The Auxiliary eqn is:-

$$m^2 + 3m + 2 = 0$$

$$\therefore m = -1, -2$$

roots are real and unequal

$\therefore$  The Complementary function:-

$$y_c = c_1 e^{-2t} + c_2 e^{-t}$$

Now:- The particular Integral:-

$$y_p = At^2 + Bt + C \quad \textcircled{iii}$$

$$y_p' = 2At + B$$

$$y_p'' = 2A$$

From eq  $\textcircled{i}$ :-

$$2A + 6At + 3B + 2At^2 + 2Bt + 2C = t^2$$

Equating like terms:—

$$2A = 1$$

$$\therefore A = \frac{1}{2}$$

$$6A + 2B = 0$$

$$\therefore B = -\frac{3}{2}$$

$$2A + 3B + 2C = 0$$

$$\therefore C = -\frac{2 \times \frac{1}{2} + 3 \times -\frac{3}{2}}{2}$$
$$= \frac{2}{4}$$

Put eq (iii) =

$$y_p = \frac{1}{2}t^2 - \frac{3}{2}t + \frac{7}{4}$$

The general soln:—

$$y = y_c + y_p$$

$$= C_1 e^{-2t} + C_2 e^{-t} + \frac{1}{2}t^2 - \frac{3}{2}t + \frac{7}{4}$$

Ans:—



$$(37) \frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y = e^{-t} + 4 \quad \text{--- (I)}$$

Consider,  $\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y = 0$  --- (II)

Let,  $y = e^{mt}$  be a trial soln on the eq (II)  $\Rightarrow$

So, The A.E. is ---

$$m^2 + 5m + 6 = 0$$

$$\therefore m = -2, -3$$

Roots are real and unequal.

$\therefore$  The C.F. is ---

$$y_c = c_1 e^{-3t} + c_2 e^{-2t}$$

Now,

The P.I. is ---

$$y_p = A e^{-t} + B \quad \text{--- (III)}$$

$$y_p' = -A e^{-t}$$

$$y_p'' = A e^{-t}$$

From eq (I)  $\Rightarrow$

$$A e^{-t} - 5A e^{-t} + 6A e^{-t} + 6B = e^{-t} + 4$$

$$2A e^{-t} + 6B = e^{-t} + 4$$

Equating like terms: ---

$$2A = 1$$

$$\therefore A = \frac{1}{2}$$

$$6B = 4$$

$$\therefore B = \frac{2}{3}$$

Put it in eq (III)  $\Rightarrow$

$$y_p = \frac{1}{2} e^{-t} + \frac{2}{3}$$

$\therefore$  The general soln: ---

$$y = y_c + y_p \\ = c_1 e^{-3t} + c_2 e^{-2t} + \frac{1}{2} e^{-t} + \frac{2}{3}$$

(139)  $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 2t + e^{-t}$  ————— (i)

Consider:-  $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 0$  ————— (ii)

Let,  $y = e^{mt}$  be a trial soln of eq (ii)  $\Rightarrow$

So, The A.E.:-

$$m^2 + 6m + 8 = 0$$

$$\therefore m = -2, -4,$$

The roots are real and unequal.

$\therefore$  The C.F.:-

$$y_c = c_1 e^{-4t} + c_2 e^{-2t}$$

Now! The P.I.:-

$$y_p = At + B + Ce^{-t} \text{ ————— (iii)}$$

$$y_p' = A - Ce^{-t}$$

$$y_p'' = +Ce^{-t}$$

From eq (i)  $\Rightarrow$

$$Ce^{-t} + 6A - 6Ce^{-t} + 8At + 8B + 8Ce^{-t} = 2t + e^{-t}$$

$$3Ce^{-t} + 8At + 6A + 8B = 2t + e^{-t}$$

Equating like terms:-

$$3C = 1 \Rightarrow C = \frac{1}{3}; 8A = 2 \Rightarrow A = \frac{1}{4}; 6A + 8B = 0$$

$$6 \times \frac{1}{4} + 8B = 0$$

$$\therefore B = -\frac{1}{4}$$

Put it in eq (iii)  $\Rightarrow$

$$y_p = \frac{1}{4}t - \frac{1}{4} + \frac{1}{3}e^{-t}$$

$\therefore$  The general soln:-

$$y = c_1 e^{-4t} + c_2 e^{-2t} + \frac{1}{4}t - \frac{1}{4} + \frac{1}{3}e^{-t}$$

Ans:-



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$$(4) \frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y = 2 \sin t \quad \text{--- (i)}$$

Consider:

$$\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y = 0 \quad \text{--- (ii)}$$

Let  $y = e^{mt}$  be a trial soln of eq (ii)

So, The AE:-

$$m^2 + 3m + 2 = 0$$

$$\therefore m = -2, -1$$

Roots are real and unequal.

$\therefore$  The CF:-

$$y_c = c_1 e^{-2t} + c_2 e^{-t}$$

Now:-

The PI:-

$$y_p = A \cos t + B \sin t \quad \text{--- (iii)}$$

$$y_p' = -A \sin t + B \cos t$$

$$y_p'' = -A \cos t - B \sin t$$

From eqn (i) :-

$$-A \cos t - B \sin t - 3A \sin t + 3B \cos t + 2A \cos t + 2B \sin t = 2 \sin t$$

$$A \cos t + B \sin t - 3A \sin t + 3B \cos t = 2 \sin t$$

Equating like terms:-

$$\sin t (B - 3A) = 2 \sin t \quad ; \quad \cos t (A + 3B) = 0$$

$$B - 3A = 2 \quad \text{--- (iv)}$$

$$A + 3B = 0 \quad \text{--- (v)}$$

(iv) + (v)  $\times 3$  :-

$$-3A + B = 2$$

$$3A + 9B = 0$$

$$10B = 2$$

$$\therefore B = \frac{1}{5}$$

$$\therefore A = -\frac{3}{5}$$



put it in eq (iii)  $\Rightarrow$

$$y_p = -\frac{3}{5} \cos t + \frac{1}{5} \sin t$$

$\therefore$  The General soln: -

$$y = y_e + y_p \\ = ce^{-2t} + eze^{-t} - \frac{3}{5} \cos t + \frac{1}{5} \sin t$$

Ans: -

$$(10) \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} + y = \cos 3t \quad \text{--- (i)}$$

$$\Rightarrow \text{Consider: - } \frac{d^2y}{dt^2} + 2 \frac{dy}{dt} + y = 0 \quad \text{--- (ii)}$$

Let  $y = e^{mt}$  be a trial soln of eq (ii)  $\Rightarrow$

The AE: -

$$m^2 + 2m + 1 = 0$$

$$\therefore m = -1, -1$$

The roots are real and equal: -

$\therefore$  The CF: -

$$y_e = (c_1 + c_2 t) e^{-t}$$

Now: The PI: -

$$y_p = A \cos 3t + B \sin 3t \quad \text{--- (iii)}$$

$$y_p' = -3A \sin 3t + 3B \cos 3t$$

$$y_p'' = -9A \cos 3t - 9B \sin 3t$$

From eq (i)  $\Rightarrow$

$$\begin{aligned} & -9A \cos 3t - 9B \sin 3t - 6A \sin 3t + 6B \cos 3t + A \cos 3t + B \sin 3t = \cos 3t \\ & -8A \cos 3t - 8B \sin 3t - 6A \sin 3t + 6B \cos 3t = \cos 3t \\ & \cos 3t (-8A + 6B) + \sin 3t (-8B - 6A) = \cos 3t \end{aligned}$$

Equating like terms:

$$-8A + 6B = 1 \quad \text{--- (iv)} ; \quad -8B - 6A = 0 \quad \text{--- (v)}$$

(iv)  $\times 4$  + (v)  $\times 3$   $\Rightarrow$

$$-32A + 24B = 4$$

$$-18A - 24B = 0$$

---

$$-50A = 4$$

$$\therefore A = -\frac{2}{25}$$

$$\therefore B = \frac{3}{50}$$

Put it in eq (ii)  $\Rightarrow$

$$y_p = -\frac{2}{25} \cos 2t + \frac{3}{50} \sin 2t$$

$\therefore$  The general eqn:

$$y = y_e + y_p$$

$$= (c_1 + c_2 t) e^{-t} + \frac{3}{50} \sin 2t - \frac{2}{25} \cos 2t$$

Ans:-

# Find the Solution of the  
given initial Value problem:-

(10)  $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 2\cos 3t$  ————— (1)  
 $y(0) = y'(0) = 0$

Consider:-  
non

$\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 8y = 0$  ————— (11)

Let,  $y = e^{mt}$  be a trial soln of eq (11)

So, The AE:-  
non

$m^2 + 6m + 8 = 0$

$\therefore m = -2, -4$

roots are real and unequal.

$\therefore$  The CF:-  
non

$y_c = c_1 e^{-2t} + c_2 e^{-4t}$

Now, The PI:-  
non

$y_p = A \cos 3t + B \sin 3t$  ————— (111)

$y_p' = -3A \sin 3t + 3B \cos 3t$

$y_p'' = -9A \cos 3t - 9B \sin 3t$

Put from eq (11):

$-9A \cos 3t - 9B \sin 3t - 8A \sin 3t + 18B \cos 3t + 8A \cos 3t + 8B \sin 3t = 2 \cos 3t$   
 $-A \cos 3t - B \sin 3t - 18A \sin 3t + 18B \cos 3t = 2 \cos 3t$   
 $\cos 3t(-A + 18B) + \sin 3t(-B - 18A) = 2 \cos 3t$



Equating like terms:-

$$\cos 3t(-A + 18B) = 2 \cos 3t; \quad -B - 18A = 0$$

$$\therefore -A + 18B = 2 \quad \text{--- (IV)}$$

$$-18A - B = 0 \quad \text{--- (V)}$$

(IV) + (V) \times 18 \Rightarrow

$$-A + 18B = 2$$

$$-324A - 18B = 0$$

$$\hline -325A = 2$$

$$\therefore A = -\frac{2}{325}$$

$$\therefore B = \frac{36}{325}$$

putting in eq (II) \Rightarrow

$$y_p = -\frac{2}{325} \cos 3t + \frac{36}{325} \sin 3t$$

\therefore The General Soln:-

$$y = c_1 e^{-4t} + c_2 e^{-2t} - \frac{2}{325} \cos 3t + \frac{36}{325} \sin 3t$$

$$y' = -4c_1 e^{-4t} - 2c_2 e^{-2t} + \frac{6}{325} \sin 3t + \frac{108}{325} \cos 3t \quad \text{--- (VI)}$$

Using  $y(0) = 0$  in eq (VI) \Rightarrow

$$c_1 + c_2 = -\frac{2}{325} \quad \text{--- (VII)}$$

Using  $y'(0) = 0$  in eq (VII) \Rightarrow

$$-4c_1 - 2c_2 = -\frac{108}{325}$$

$$\therefore 4c_1 + 2c_2 = \frac{108}{325} \quad \text{--- (ix)}$$

$$\therefore c_1 = \frac{4}{25}; \quad c_2 = -\frac{2}{13}$$

\therefore The General Soln:-

$$y = \frac{4}{25} e^{-4t} - \frac{2}{13} e^{-2t} - \frac{2}{325} \cos 3t + \frac{36}{325} \sin 3t$$

Ans.

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Q.  $\frac{d^2y}{dt^2} + 4y = -\cos \frac{t}{2}$  ————— (i)

Consider:  $\frac{d^2y}{dt^2} + 4y = 0$  ————— (ii)

Let,  $y = e^{mt}$  be a trial solution of eq (ii)

So, The AE:—

$$m^2 + 4 = 0$$

$$\therefore m = \pm 2i$$

The roots are complex conjugate

$\therefore$  The IP:—

$$y_p = A \cos \frac{t}{2} + B \sin \frac{t}{2} \text{ ————— (iii)}$$

$$y_p' = -\frac{1}{2}A \sin \frac{t}{2} + \frac{1}{2}B \cos \frac{t}{2}$$

$$y_p'' = -\frac{1}{4}A \cos \frac{t}{2} - \frac{1}{4}B \sin \frac{t}{2}$$

From: eq (i)

$$-\frac{1}{4}A \cos \frac{t}{2} - \frac{1}{4}B \sin \frac{t}{2} + 4A \cos \frac{t}{2} + 4B \sin \frac{t}{2} = -\cos \frac{t}{2}$$
$$\cos \frac{t}{2} \left( -\frac{1}{4}A + 4A \right) + \sin \frac{t}{2} \left( -\frac{1}{4}B + 4B \right) = -\cos \frac{t}{2}$$

Equating like terms:—

$$-\frac{1}{4}A + 4A = -1 \quad ; \quad -\frac{1}{4}B + 4B = 0$$
$$A = -\frac{1}{(-\frac{1}{4} + 4)} = \frac{4}{17} \quad \therefore B = 0$$



put it in eq (i) =

$$-DA \cos 2t - DB \sin 2t + 4A \cos 2t + 4B \sin 2t = \sin 2t$$

$$-5A \cos 2t - 5B \sin 2t = \sin 2t$$

Equating like terms:-

$$-5B = 1 \quad ; \quad -5A = 0$$

$$\therefore B = -\frac{1}{5} \quad \therefore A = 0$$

put it in eq (ii) =

$$Y_p = -\frac{1}{5} \sin 2t$$

So, The general soln:-

$$Y = Y_c + Y_p$$

$$Y = C_1 \cos 2t + C_2 \sin 2t - \frac{1}{5} \sin 2t \quad \text{--- (iv)}$$

$$Y' = -2C_1 \sin 2t + 2C_2 \cos 2t - \frac{2}{5} \cos 2t \quad \text{--- (v)}$$

using  $Y(0) = 2$  in eqn (iv) =

$$C_1 = 2$$

using  $Y'(0) = 0$  in eqn (v) =

$$2C_2 = \frac{2}{5}$$

$$\therefore C_2 = \frac{1}{5}$$

$\therefore$  The general soln is:-

$$Y = 2 \cos 2t + \frac{1}{5} \sin 2t - \frac{1}{5} \sin 2t$$

Ans:-

Put in eqn (ii)  $\Rightarrow$

$$y_p = \frac{4}{17} \cos \frac{t}{2}$$

$\therefore$  The general soln:—

$$y = y_e + y_p$$

$$= c_1 \cos 2t + c_2 \sin 2t + \frac{4}{17} \cos \frac{t}{2}$$

Ans:—

(14)  $\frac{d^2 y}{dt^2} + 4y = 8 \sin 3t$

$$y(0) = 2, y'(0) = 0$$

Consider:—  $\frac{d^2 y}{dt^2} + 4y = 0$

Let:  $y = e^{mt}$  be a trial soln of eq (ii)  $\Rightarrow$

$\therefore$  The AE:—

$$m^2 + 4 = 0$$

$$\therefore m = \pm 2i$$

The roots are

$\therefore$  The CF:—

$$y_e = (c_1 \cos 2t + c_2 \sin 2t)$$

Complex Conjugate

Now, The PI:—

$$y_p = A \cos 3t + B \sin 3t$$

$$y_p' = -3A \sin 3t + 3B \cos 3t$$

$$y_p'' = -9A \cos 3t - 9B \sin 3t$$

#System of First  
Order ODE



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# Find the general Soln of the following D.E

$$\textcircled{3} \frac{dy}{dt} = -v \quad \text{--- (i)}$$

$$\frac{dv}{dt} = y$$

(Using the Differential operation  $\frac{d}{dt} \equiv D$ )

$$\therefore Dy = -v \quad \text{--- (ii)}$$

$$Dv = y \quad \text{--- (iii)}$$

Now:  $\textcircled{ii} \times D + \textcircled{iii} \Rightarrow$

$$D^2y = -Dv$$

$$y = Dv$$

$$D^2y + y = 0$$

$$\therefore \frac{d^2y}{dt^2} + y = 0 \quad \text{--- (iv)}$$

Let,  $y = e^{mt}$  be a trial soln of  $\textcircled{iv} \Rightarrow$

so, The Auxiliary Eqn:—

$$m^2 + 1 = 0$$

$$\therefore m = \pm i$$

roots are

Complex Conjugate

$\therefore$  The general Soln:—

$$y(t) = c_1 \cos t + c_2 \sin t$$

From eqn (i)  $\Rightarrow$

$$\frac{d}{dt} \{ c_1 \cos t + c_2 \sin t \} = -v$$

$$-c_1 \sin t + c_2 \cos t = -v$$

$$\therefore v(t) = c_1 \sin t - c_2 \cos t$$

(i)

(iii)

$$\frac{dx}{dt} = x + 2y$$

(ii)

$$\frac{dy}{dt} = -y$$

Let

$$\frac{d}{dt} \equiv D$$

So

$$Dx = x + 2y$$

$$\therefore Dx - x - 2y = 0 \implies x(D-1) - 2y = 0 \quad \text{--- (1)}$$

Again:-

$$Dy = -y$$

$$\therefore Dy + y = 0$$

$$y(D+1) = 0 \quad \text{--- (2)}$$

$$\textcircled{1} \times (D+1) + \textcircled{2} \times 2 \Rightarrow$$

$$x(D^2-1) - 2(D+1)y = 0$$

$$0 + 2(D+1)y = 0$$

$$x(D^2-1) = 0$$

$$xD^2 - x = 0$$

$$\therefore \frac{d^2x}{dt^2} - x = 0$$

Let

$y = e^{mt}$  be a trial soln:-

So The AEqn:-

$$m^2 - 1 = 0$$

$$\therefore m = \pm 1$$

roots are real and not equal

$$\therefore x(t) = c_1 e^t + c_2 e^{-t}$$

$\therefore$  Put  $x$  in eq (1) :-

$$\frac{d}{dt}(c_1 e^t + c_2 e^{-t}) = x + 2y$$

$$\begin{aligned} \frac{d}{dt}(c_1 e^t + c_2 e^{-t}) &= c_1 e^t - c_2 e^{-t} + 2y \\ -c_1 e^t + c_2 e^{-t} - c_1 e^t + c_2 e^{-t} &= 2y \\ \therefore 2y &= -2c_1 e^t \end{aligned}$$

$$\therefore y = -c_1 e^t$$



$$\text{viii) } \frac{dx}{dt} = x - 2y \quad \text{--- (i)}$$

$$\frac{dy}{dt} = -y \quad \text{--- (ii)}$$

$$\text{Let } D \equiv \frac{d}{dt}$$

$$\therefore Dx = x - 2y$$

$$\Rightarrow Dx - x = -2y$$

$$\Rightarrow x(D-1) + 2y = 0 \quad \text{--- (iii)}$$

Again

$$Dy = -y$$

$$\Rightarrow Dy + y = 0$$

$$\therefore y(D+1) = 0 \quad \text{--- (iv)}$$

$$\text{(iii)} \times (D+1) - \text{(iv)} \times 2 =$$

$$x(D^2-1) + 2y(D+1) = 0$$

$$-2y(D+1) = 0$$

$$x(D^2-1) = 0$$

$$\therefore xD^2 - x = 0$$

$$\therefore \frac{d^2x}{dt^2} - x = 0 \quad \text{--- (v)}$$

Let  $y = e^{mt}$  be trial soln of eqn (v)

So The AE:-

$$m^2 - 1 = 0$$

$$\therefore m = \pm 1$$

Roots are real and not equal.

$$\therefore x(t) = c_1 e^{-t} + c_2 e^t$$

put the value of x in eq (i)

$$\frac{d}{dt}(c_1 e^{-t} + c_2 e^t) = c_1 e^{-t} + c_2 e^t - 2y$$

$$-c_1 e^{-t} + c_2 e^t - c_1 e^{-t} - c_2 e^t = -2y$$

$$\therefore -2y = -2c_1 e^{-t}$$

$$\therefore y = c_1 e^{-t}$$

$$\therefore y(t) = c_1 e^{-t}$$



100 Days Camp



# # Application of Separation of Variables

$$\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = A$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 14 \\ 32 \\ 32 \end{pmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 14 \\ 32 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{pmatrix} 14 \\ 32 \\ 32 \end{pmatrix}$$

$$(11) \rightarrow 0 = 14 + 32 + 32$$

$$(12) \rightarrow 0 = 14 + 32 + 32$$

$$(13) \rightarrow 0 = 14 + 32 + 32$$

$$(14) \rightarrow 0 = 14 + 32 + 32$$

$$(15) \rightarrow 0 = 14 + 32 + 32$$

$$(16) \rightarrow 0 = 14 + 32 + 32$$

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Example: —  $\frac{dx}{dt} = -x + y$  — (i)

$\frac{dy}{dt} = -3x - 5y$  — (iii)

Let:  $D = \frac{d}{dt}$

$\therefore Dx = -x + y$

$\Rightarrow Dx + x - y = 0$

$\Rightarrow x(D+1) - y = 0$  — (ii)

Again:  $Dy = -3x - 5y$

$\Rightarrow Dy + 5y + 3x = 0$

$\Rightarrow y(D+5) + 3x = 0$  — (iv)

Now: (ii)  $\times (D+5)$  — (v)

$x(D+5)(D+1) - y(D+5) = 0$

$3x + y(D+5) = 0$

$x(D+5)(D+1) + 3x = 0$

$\therefore D^2x + 6Dx + 8x = 0$

$\Rightarrow \frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 8x = 0$

Let  $y = e^{mt}$  be a trial soln: —

'So, The AE: —

$m^2 + 6m + 8 = 0$

$\therefore m = -4, -2$

roots are real and not equal

So:  $x(t) = c_1 e^{-4t} + c_2 e^{-2t}$

Put the value of  $x$  in eq (i)

$\frac{d}{dt}(c_1 e^{-4t} + c_2 e^{-2t}) = -c_1 e^{-4t} - c_2 e^{-2t} + y$   
 $-4c_1 e^{-4t} - 2c_2 e^{-2t} = -c_1 e^{-4t} - c_2 e^{-2t} + y$

$\therefore y(t) = -3c_1 e^{-4t} - c_2 e^{-2t}$



(26)  $A = \begin{pmatrix} -2 & -1 \\ 2 & -5 \end{pmatrix}$

Let  $\begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} -2 & -1 \\ 2 & -5 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}$

Now  $\begin{bmatrix} -2 & -1 \\ 2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2x - y \\ 2x - 5y \end{bmatrix}$

$\therefore \begin{pmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \end{pmatrix} = \begin{pmatrix} -2x - y \\ 2x - 5y \end{pmatrix} \therefore \frac{dx}{dt} = -2x - y \quad \text{--- (I)}$

$\frac{dy}{dt} = 2x - 5y \quad \text{--- (II)}$

Again:- Let  $D \equiv \frac{d}{dt}$

$\therefore Dx + 2x + y = 0$   
 $x(D+2) + y = 0 \quad \text{--- (III)}$

Again  $Dy + 5y - 2x = 0$   
 $y(D+5) - 2x = 0 \quad \text{--- (IV)}$

(III)  $\times 2 +$  (IV)  $\times (D+2) \Rightarrow$

$2x(D+2) + 2y = 0$   
 $-2x(D+2) + y(D+5)(D+2) = 0$

$y(D+5)(D+2) + 2y = 0$

$D^2y + 7Dy + 12y = 0$

$\therefore \frac{d^2y}{dt^2} + 7\frac{dy}{dt} + 12y = 0$

Let  $y = e^{mt}$  be a trial soln:-

So The AE:-

$m^2 + 7m + 12 = 0$

$\therefore m = -4, -3$

Roots are real and not equal.

So  $y(t) = c_1 e^{-4t} + c_2 e^{-3t}$

Put the value of  $y$  in eq (I)

$\frac{d}{dt}(c_1 e^{-4t} + c_2 e^{-3t}) = 2x + c_1 e^{-4t} + c_2 e^{-3t}$

$-4c_1 e^{-4t} - 3c_2 e^{-3t} = 2x + c_1 e^{-4t} + c_2 e^{-3t}$

$2x = c_1 e^{-4t} + 2c_2 e^{-3t}$

$\therefore x(t) = \frac{1}{2} c_1 e^{-4t} + c_2 e^{-3t}$

40:- Consider the following very simple model of blood cholesterol levels based on the fact that cholesterol is manufactured by the body for use in the construction of cell walls and is absorbed from foods containing cholesterol. Let  $C(t)$  be the amount of cholesterol in the blood of a particular person at time  $t$ .

$$\frac{dc}{dt} = k_1(N - c) + k_2E$$

Where,  $N$  = the person's natural cholesterol level.

$k_1$  = production parameter

$E$  = daily rate at which cholesterol is eaten and

$k_2$  = absorption parameter.

- (a) Suppose  $N = 200$ ,  $k_1 = 0.1$ ,  $k_2 = 0.1$ ,  $E = 400$ , and  $c(0) = 150$ . What will the person's cholesterol level be after 2 days on this diet?
- (b) With the initial condition as above, what will the person's cholesterol level be after 5 days on this diet?



After 2 Days  $t=2$

$$\begin{aligned}\therefore C(t) &= 600 - 10 \times A' e^{-0.1t} \\ &= 600 - 10 \times 45 \cdot e^{-0.1 \times 2} \\ &= 231.57\end{aligned}$$

After 5 Days  $t=5$

$$\begin{aligned}C(t) &= 600 - 10 A' e^{-0.1t} \\ &= 600 - 10 \times 45 \times e^{-0.1 \times 5} \\ &= 327.06\end{aligned}$$

① The population of a country be double within 30 years. How many years it will take triple. Let the increase rate is proportional to the population remaining.

$\Rightarrow$  Let  $P$  = remaining population

$\therefore$  According to the question:

$$\frac{dP}{dt} \propto P$$

$$\Rightarrow \frac{dP}{dt} = KP$$

$$\Rightarrow \frac{dP}{P} = K dt$$

$$\Rightarrow \int \frac{dP}{P} = K \int dt$$

$$\therefore \ln(P) = Kt + \ln c$$



(a)

In given:

$$\frac{dc}{dt} = k_1(N-c) + k_2E$$

$$\Rightarrow \frac{dc}{dt} = 0.1(200-c) + 0.1 \times 400$$

$$\Rightarrow \frac{dc}{dt} = 20 - 0.1c + 40$$

$$\Rightarrow \frac{dc}{60 - 0.1c} = dt$$

$$\Rightarrow \int \frac{dc}{60 - 0.1c} = \int dt$$

$$\Rightarrow -\frac{1}{0.1} \ln(60 - 0.1c) = t + C'$$

$$\Rightarrow \ln(60 - 0.1c) = -0.1t - 0.1C' \quad [A = -0.1C']$$

$$\Rightarrow \ln(60 - 0.1c) = -0.1t + A$$

$$\Rightarrow 60 - 0.1c = e^{-0.1t} \cdot e^A \quad [A' = e^A]$$

$$\Rightarrow 60 - A'e^{-0.1t} = 0.1c$$

$$\therefore c(t) = 600 - 10A'e^{-0.1t} \quad \text{--- (1)}$$

∴ Applying initial condition:  $c(0) = 150$

$$t=0, c=150$$

$$150 = 600 - 10A'e^{-0.1 \times 0}$$

$$\therefore A' = \frac{600 - 150}{10}$$

$$= 45$$

$$\Rightarrow \ln\left(\frac{P}{c}\right) = kt$$

$$\Rightarrow \frac{P}{c} = e^{kt}$$

$$\Rightarrow \boxed{P = ce^{kt}} \quad \text{--- (11)}$$

When,  $t=0$ ,  $P=P_0$

$$\therefore P_0 = ce^{k \cdot 0}$$

$$\boxed{c = P_0}$$

When:-  $t=30$  years  $P=2P_0$

$$\therefore 2P_0 = P_0 e^{k \cdot 30}$$

$$20k = \ln 2$$

$$\boxed{\therefore k = 0.02310}$$

Now:- When:-  $t=?$   $P=3P_0$

$$3P_0 = P_0 e^{kt}$$

$$e^{kt} = 3$$

$$t = \frac{\ln 3}{0.02310}$$

$$= 47.54$$

$$\approx 48 \text{ years.}$$

② Newton's law of cooling state that the rate of change of temperature of a body is proportional to the difference between the temperature of body and that of the surrounding medium. The room temperature  $70^{\circ}\text{F}$ , When a cake is removed from the oven, its temperature measured at  $300^{\circ}\text{F}$  Three min later its temperature is  $200^{\circ}\text{F}$ . How long will it take for the cake to cool off a temperature of  $75^{\circ}\text{F}$ ?

Let  $T_B$  = The body temperature.

$T_S$  = The Surrounding temperature.

According to the question:-

$$\frac{dT_B}{dt} \propto (T_B - T_S)$$

$$\Rightarrow \frac{dT_B}{dt} = K(T_B - T_S)$$

$$\Rightarrow \frac{dT_B}{dt} = K(T_B - 70^{\circ})$$

$$\Rightarrow \frac{dT_B}{T_B - 70} = K \cdot dt$$

$$\Rightarrow \int \frac{dT_B}{T_B - 70} = \int K \cdot dt$$

$$\Rightarrow \ln(T_B - 70) = Kt + \ln C$$

$$\Rightarrow \ln\left(\frac{T_B - 70}{C}\right) = Kt$$



$$\Rightarrow \frac{T_B - T_0}{C} = e^{kt}$$

$$\Rightarrow \boxed{T_B - T_0 = Ce^{kt}}$$

When:  $t = 0, T_B = 300^\circ \text{F}$

$$\therefore 300 - 70 = Ce^0$$

$$\therefore \boxed{C = 230}$$

When:  $t = 3 \text{ min}, T_B = 200^\circ \text{F}$

$$\therefore 200 - 70 = 230e^{k \cdot 3}$$

$$k = \frac{\ln\left(\frac{130}{230}\right)}{3}$$

$$= -0.10$$

Now, when  $T_B = 75, t = ?$

$$75 - 70 = 230e^{-0.10t}$$

$$e^{-0.10t} = \frac{5}{230}$$

$$t = \frac{\ln\left(\frac{5}{230}\right)}{-0.10}$$

$$= 20.25 \text{ min}$$

$$\approx \underline{20 \text{ min}}$$